Class 10 Physics - Solved Exercises (PCTB) - Units 10-18

Unit 10: Simple Harmonic Motion and Waves

Multiple Choice Questions

i. Which of the following is an example of simple harmonic motion?

✓ (a) the motion of simple pendulum

(b) the motion of ceiling fan

(c) the spinning of the Earth on its axis

(d) a bouncing ball on a floor

ii. If the mass of the bob of a pendulum is increased by a factor of 3, the period of the pendulum's motion will

(a) be increased by a factor of 2

✓ (b) remain the same

(c) be decreased by a factor of 2

(d) be decreased by a factor of 4

iii. Which of the following devices can be used to produce both transverse and longitudinal waves?

(a) a string

(b) a ripple tank

✓ (c) a helical spring (slinky)

(d) a tuning fork

iv. Waves transfer

✓ (a) energy

(b) frequency

(c) wavelength

(d) velocity

v. Which of the following is a method of energy transfer?

(a) conduction

(b) radiation

(c) wave motion

✓ (d) all of these

vi. In a vacuum, all electromagnetic waves have the same

✓ (a) speed

(b) frequency

(c) amplitude

(d) wavelength

vii. A large ripple tank with a vibrator working at a frequency of 30 Hz produces 25 complete waves in a distance of 50 cm. The velocity of the wave is

(a) 53 cm/s

✓ (b) 60 cm/s

(c) 750 cm/s

(d) 1500 cm/s

Wavelength = 50 cm / 25 = 2 cm. Velocity v = f x wavelength = 30 x 2 = 60 cm/s.

viii. Which of the following characteristics of a wave is independent of the others?

(a) speed

(b) frequency

✓ (c) amplitude

(d) wavelength

ix. The relation between v, f and wavelength of a wave is

(a) v f = wavelength

✓ (b) f x wavelength = v

(c) v x wavelength = f

(d) v = wavelength / f

Review Questions

10.1. What is simple harmonic motion? What are the necessary conditions for a body to execute simple harmonic motion?

Simple harmonic motion is the to-and-fro motion of a body about a mean position in which its acceleration is always directed towards the mean position and is proportional to its displacement from it. The necessary conditions are: a restoring force must act on the body, this force must always point towards the mean position, and its magnitude must increase as the displacement from the mean position increases.

10.2. Think of several examples of motion in everyday life that are simple harmonic.

Everyday examples of simple harmonic motion are the swing of a simple

pendulum in a clock, the vibrations of a guitar string, the motion of a mass hanging from a spring, the motion of a ball rolling in a bowl, and the vibrations of a tuning fork. In each case the body moves to and fro about a fixed mean position under a restoring force.

10.3. What are damped oscillations? How does damping progressively reduce the amplitude of oscillation?

Damped oscillations are oscillations whose amplitude decreases gradually with time because of resistive forces such as friction and air resistance. In every vibration some energy is used up in overcoming these resistances, so less energy remains for the next vibration. As the energy of oscillation falls, the amplitude also falls, until the motion finally stops.

10.4. How can you define the term wave? Elaborate the difference between mechanical and electromagnetic waves. Give examples of each.

A wave is a disturbance that transfers energy from one place to another without transferring matter. Mechanical waves need a material medium for their travel, for example sound waves and water waves. Electromagnetic waves do not need any medium and can travel through vacuum, for example light waves, radio waves and X-rays.

10.5. Distinguish between longitudinal and transverse waves with suitable examples.

In a transverse wave the particles of the medium vibrate at right angles to the direction of travel of the wave, for example waves on a rope and light waves. In a longitudinal wave the particles vibrate along the direction of travel, forming compressions and rarefactions, for example

sound waves and waves in a slinky spring.

10.6. Draw a transverse wave with an amplitude of 2 cm and a wavelength of 4 cm. Label a crest and trough on the wave.

Draw a smooth sine-shaped wave on a horizontal axis. Mark the highest points as crests and the lowest points as troughs. The vertical distance from the middle line to a crest is 2 cm, which is the amplitude. The horizontal distance between two neighbouring crests is 4 cm, which is the wavelength.

10.7. Derive a relationship between velocity, frequency and wavelength of a wave. Write a formula relating velocity of a wave to its time period and wavelength.

In one time period T a wave travels one wavelength, so its velocity is v = wavelength / T. Since frequency f = 1 / T, this becomes v = f x wavelength. In terms of time period alone, the relation is v = wavelength / T, where wavelength is the distance between two neighbouring crests.

10.8. Waves are the means of energy transfer without transfer of matter. Justify this statement with the help of a simple experiment.

Place a cork on the surface of water in a ripple tank and produce waves at one end. The cork moves up and down at its own place as each wave passes, but it does not travel forward with the wave. This shows that the wave carries energy forward and makes the cork vibrate, while the water particles themselves do not travel with the wave.

10.9. Explain the following properties of waves with reference to ripple tank experiment: a. Reflection b. Refraction c. Diffraction

Reflection: when straight waves in a ripple tank strike a flat barrier they bounce back, and the angle of reflection equals the angle of incidence. Refraction: when waves pass from deep to shallow water their speed and wavelength decrease and their direction bends, while frequency stays the same. Diffraction: when waves pass through a narrow gap or round an edge they spread out into the shadow region behind the obstacle.

10.10. Does increasing the frequency of a wave also increase its wavelength? If not, how are these quantities related?

No, increasing the frequency does not increase the wavelength. For a wave travelling at a fixed speed, v = f x wavelength, so frequency and wavelength are inversely related. When frequency increases, wavelength decreases in the same proportion, and their product remains equal to the wave speed.

Conceptual Questions

10.1. If the length of a simple pendulum is doubled, what will be the change in its time period?

The time period of a simple pendulum is T = 2 x pi x square root (length / g), so T is proportional to the square root of the length. Doubling the length multiplies T by the square root of 2, that is about 1.41 times. The new time period is therefore about 1.41 times the old time period.

10.2. A ball is dropped from a certain height onto the floor and keeps bouncing. Is the motion of the ball simple harmonic? Explain.

No, the bouncing ball is not in simple harmonic motion. Although it

moves up and down repeatedly, its acceleration is not proportional to its displacement from a mean position and is not always directed towards that position. Between bounces it falls freely under gravity, and the motion is interrupted at each impact, so the conditions for simple harmonic motion are not met.

10.3. A student performed two experiments with a simple pendulum using bobs of different masses, keeping other parameters constant. The time period did not change. Why?

The time period of a simple pendulum, T = 2 x pi x square root (length / g), does not contain the mass of the bob. A heavier bob experiences a larger restoring force, but it also has larger inertia, and the two effects cancel exactly. The time period therefore depends only on the length and on g, not on the mass.

10.4. What types of waves do not require any material medium for their propagation?

Electromagnetic waves do not require any material medium. They are produced by vibrating electric charges and consist of oscillating electric and magnetic fields, so they can travel through vacuum. Examples are light, radio waves, microwaves, infrared rays, ultraviolet rays, X-rays and gamma rays.

10.5. Plane waves in the ripple tank undergo refraction when they move from deep to shallow water. What change occurs in the speed of the waves?

When waves move from deep water into shallow water their speed decreases. The wavelength also decreases in the same proportion, while

the frequency remains unchanged. Because the speed changes, the direction of the waves bends at the boundary, which is refraction.

Numerical Problems

10.1. The time period of a simple pendulum is 2 s. What will be its length on the Earth? What will be its length on the Moon if g on Moon = g on Earth / 6, where g on Earth = 10 m/s^2?

Step 1: Use T = 2 x pi x square root (l / g), so l = g x T^2 / (4 x pi^2).

Step 2: On Earth: l = 10 x (2)^2 / (4 x 3.14^2) = 40 / 39.48 = 1.01 m, about 1.02 m.

Step 3: On the Moon g = 10 / 6 = 1.67 m/s^2, so l = 1.67 x 4 / 39.48 = 0.17 m.

Final answer: Length on Earth = 1.02 m; length on Moon = 0.17 m.

10.2. A pendulum of length 0.99 m is taken to the Moon by an astronaut. The period of the pendulum is 4.9 s. What is the value of g on the surface of the Moon?

Step 1: From T = 2 x pi x square root (l / g), we get g = 4 x pi^2 x l / T^2.

Step 2: Substitute l = 0.99 m and T = 4.9 s: g = 4 x (3.14)^2 x 0.99 / (4.9)^2.

Step 3: g = 39.48 x 0.99 / 24.01 = 39.09 / 24.01 = 1.63 m/s^2.

Final answer: g on the Moon = 1.63 m/s^2.

10.3. Find the time periods of a simple pendulum of 1 metre length, placed on Earth and on Moon. The value of g on the surface of Moon is 1/6 of its value on Earth, where g on Earth is 10 m/s^2.

Step 1: Use T = 2 x pi x square root (l / g) with l = 1 m.

Step 2: On Earth: T = 2 x 3.14 x square root (1 / 10) = 6.28 x 0.316 = 1.99 s, about 2 s.

Step 3: On the Moon g = 10 / 6 = 1.67 m/s^2, so T = 6.28 x square root (1 / 1.67) = 6.28 x 0.774 = 4.86 s, about 4.9 s.

Final answer: Time period on Earth = 2 s; on Moon = 4.9 s.

10.4. A simple pendulum completes one vibration in two seconds. Calculate its length, when g = 10.0 m/s^2.

Step 1: Time period T = 2 s and l = g x T^2 / (4 x pi^2).

Step 2: l = 10 x (2)^2 / (4 x 3.14^2) = 40 / 39.48.

Step 3: l = 1.01 m, about 1.02 m.

Final answer: Length = 1.02 m.

10.5. If 100 waves pass through a point of a medium in 20 seconds, what is the frequency and the time period of the wave? If its wavelength is 6 cm, calculate the wave speed.

Step 1: Frequency f = number of waves / time = 100 / 20 = 5 Hz.

Step 2: Time period T = 1 / f = 1 / 5 = 0.2 s.

Step 3: Wavelength = 6 cm = 0.06 m. Wave speed v = f x wavelength = 5 x 0.06 = 0.3 m/s.

Final answer: Frequency = 5 Hz, time period = 0.2 s, wave speed = 0.3 m/s.

10.6. A wooden bar vibrating into the water surface in a ripple tank has a frequency of 12 Hz. The resulting wave has a wavelength of 3 cm. What is the speed of the wave?

Step 1: Wavelength = 3 cm = 0.03 m and frequency f = 12 Hz.

Step 2: Wave speed v = f x wavelength = 12 x 0.03.

Step 3: v = 0.36 m/s.

Final answer: Speed of the wave = 0.36 m/s.

10.7. A transverse wave produced on a spring has a frequency of 190 Hz and travels along the length of the spring of 90 m, in 0.5 s. (a) What is the period of the wave? (b) What is the speed of the wave? (c) What is the wavelength of the wave?

Step 1: Period T = 1 / f = 1 / 190 = 0.0053 s, about 0.01 s to one significant figure.

Step 2: Speed v = distance / time = 90 / 0.5 = 180 m/s.

Step 3: Wavelength = v / f = 180 / 190 = 0.95 m.

Final answer: Period = 0.01 s, speed = 180 m/s, wavelength = 0.95 m.

10.8. Water waves in a shallow dish are 6.0 cm long. At one point, the water moves up and down at a rate of 4.8 oscillations per second. (a) What is the speed of the water waves? (b) What is the period of the water waves?

Step 1: Frequency f = 4.8 Hz and wavelength = 6.0 cm = 0.06 m.

Step 2: Speed v = f x wavelength = 4.8 x 0.06 = 0.288 m/s, about 0.29

m/s.

Step 3: Period T = 1 / f = 1 / 4.8 = 0.208 s, about 0.21 s.

Final answer: Speed = 0.29 m/s, period = 0.21 s.

10.9. At one end of a ripple tank 80 cm across, a 5 Hz vibrator produces waves whose wavelength is 40 mm. Find the time the waves need to cross the tank.

Step 1: Wavelength = 40 mm = 0.04 m and frequency f = 5 Hz.

Step 2: Wave speed v = f x wavelength = 5 x 0.04 = 0.2 m/s.

Step 3: Distance = 80 cm = 0.8 m. Time = distance / speed = 0.8 / 0.2 = 4 s.

Final answer: Time to cross the tank = 4 s.

10.10. What is the wavelength of the radiowaves transmitted by an FM station at 90 MHz? Where 1 M = 10^6, and speed of radiowave is 3 x 10^8 m/s.

Step 1: Frequency f = 90 MHz = 90 x 10^6 Hz = 9 x 10^7 Hz.

Step 2: Wavelength = speed / frequency = (3 x 10^8) / (9 x 10^7).

Step 3: Wavelength = 3.33 m.

Final answer: Wavelength = 3.33 m.

Unit 11: Sound

Multiple Choice Questions

i. Which is an example of a longitudinal wave?

✓ (a) sound wave

(b) light wave

(c) radiowave

(d) water wave

ii. How does sound travel from its source to your ear?

✓ (a) by changes in air pressure

(b) by vibrations in wires or strings

(c) by electromagnetic wave

(d) by infrared waves

iii. Which form of energy is sound?

(a) electrical

✓ (b) mechanical

(c) thermal

(d) chemical

iv. Astronauts in space need to communicate with each other by radio links because

(a) sound waves travel very slowly in space

(b) sound waves travel very fast in space

✓ (c) sound waves cannot travel in space

(d) sound waves have low frequency in space

v. The loudness of a sound is most closely related to its

(a) frequency

(b) period

(c) wavelength

✓ (d) amplitude

vi. For a normal person, audible frequency range for sound wave lies between

(a) 10 Hz and 10 kHz

✓ (b) 20 Hz and 20 kHz

(c) 25 Hz and 25 kHz

(d) 30 Hz and 30 kHz

vii. When the frequency of a sound wave is increased, which of the following will decrease? i. wavelength ii. period iii. amplitude

(a) i only

(b) iii only

✓ (c) i and ii only

(d) i and iii only

At a fixed speed, wavelength = speed / frequency and period = 1 / frequency, so both wavelength and period decrease when frequency increases. Amplitude is independent of frequency.

Review Questions

11.1. What is the necessary condition for the production of sound?

The necessary condition for the production of sound is a vibrating body. Whenever a body vibrates, it disturbs the particles of the medium around it, and these vibrations travel outwards as sound waves. Without vibration, no sound is produced.

11.2. What is the effect of the medium on the speed of sound? In which medium does sound travel faster: air, solid or liquid? Justify your answer.

The speed of sound depends on the medium: it is greatest in solids, less in liquids and least in gases. This is because particles in a solid are packed closely and are strongly linked, so a vibration is passed on very quickly from particle to particle. In air the particles are far apart, so the vibration is transferred more slowly.

11.3. How can you prove the mechanical nature of sound by a simple experiment?

Place an electric bell inside a glass bell jar connected to a vacuum pump and ring the bell. As air is pumped out, the sound becomes fainter and finally cannot be heard, although the bell is still ringing. This proves that sound needs a material medium and is therefore a mechanical wave.

11.4. What do you understand by the longitudinal wave? Describe the longitudinal nature of sound waves.

A longitudinal wave is a wave in which the particles of the medium vibrate along the direction of travel of the wave. Sound is a longitudinal wave because a vibrating source produces alternate compressions,

where air particles are crowded together, and rarefactions, where they are spread apart, and these regions travel forward through the air.

11.5. Sound is a form of wave. List at least three reasons to support the idea that sound is a wave.

Sound is a wave because it shows reflection, as in an echo; it shows refraction, as sound bends when passing between layers of air at different temperatures; and it shows diffraction, as sound is heard round corners although the source cannot be seen. It also carries energy without carrying matter, which is the defining property of a wave.

11.6. We know that waves manifest phenomenon of reflection, refraction and diffraction. Does sound also manifest these characteristics?

Yes, sound shows all three properties. Reflection of sound is heard as an echo when sound strikes a distant wall. Refraction of sound occurs when sound passes from one layer of air to another of different temperature and bends. Diffraction of sound is observed when sound spreads round a corner or through a doorway into a room.

11.7. What is the difference between the loudness and intensity of sound? Derive the relationship between the two.

Intensity is a physical quantity: the sound energy passing per second through unit area, measured in watts per square metre. Loudness is the sensation felt by the ear and depends on intensity and on the sensitivity of the ear. Loudness L is related to intensity I by $L = 10 \times \log (I / I0)$, where I0 is the threshold intensity of hearing, so loudness grows only logarithmically with intensity.

11.8. On what factors does the loudness of sound depend?

Loudness depends on the amplitude of vibration of the source: larger amplitude gives louder sound. It also depends on the distance from the source, because intensity falls as distance increases. It further depends on the area of the vibrating surface, the density of the medium, and the sensitivity of the listener's ear.

11.9. What do you mean by the term intensity level of the sound? Name and define the unit of intensity level of sound.

Intensity level is a measure of sound intensity on a logarithmic scale, defined as 10 times the logarithm of the ratio of the intensity to the threshold intensity. Its unit is the decibel (dB). One decibel is one-tenth of a bel, and the bel is defined so that an intensity ten times the threshold gives an intensity level of 1 bel, or 10 dB.

11.10. What are the units of loudness? Why do we use logarithmic scale to describe the range of the sound intensities we hear?

Loudness is measured in decibels (dB), and also in phons for the sensation of loudness. We use a logarithmic scale because the human ear responds to an enormous range of intensities, from 10^-12 W/m^2 to about 1 W/m^2, a range of a trillion times. A logarithmic scale compresses this huge range into manageable numbers from 0 dB to about 120 dB.

11.11. What is difference between frequency and pitch? Describe their relationship graphically.

Frequency is a physical quantity: the number of vibrations per second, measured in hertz. Pitch is the sensation by which the ear judges a sound

as high or low. Pitch increases as frequency increases, roughly in a logarithmic way. On a graph of pitch against frequency, the curve rises steeply at low frequencies and flattens at high frequencies, showing that equal ratios of frequency give equal steps in pitch.

11.12. Describe the effect of change in amplitude on loudness and the effect of change in frequency on pitch of sound.

Increasing the amplitude of a sound wave increases its loudness, because a larger amplitude carries more energy and produces a stronger sensation in the ear. Increasing the frequency increases the pitch, so the sound is heard as higher or shriller. Amplitude does not change pitch, and frequency does not by itself change loudness.

11.13. If the pitch of sound is increased, what are the changes in the following? a. the frequency b. the wavelength c. the wave velocity d. the amplitude of the wave

When pitch increases: (a) the frequency increases, because pitch depends on frequency. (b) The wavelength decreases, because wavelength = speed / frequency at a fixed speed. (c) The wave velocity stays the same, because it depends on the medium, not on the source. (d) The amplitude does not change unless the source is struck harder, so it remains unchanged.

11.14. If we clap or speak in front of a building while standing at a particular distance, we rehear our sound after sometime. Can you explain how does this happen?

This is an echo, caused by reflection of sound. The sound travels to the building, reflects from its wall, and returns to the ear after a short time.

The time delay equals twice the distance to the building divided by the speed of sound. An echo is heard as a distinct sound only if the reflecting surface is at least about 17 metres away.

11.15. What is the audible frequency range for human ear? Does this range vary with the age of people? Explain.

The audible frequency range for a normal human ear is from 20 Hz to 20,000 Hz. Yes, this range changes with age: children can hear frequencies near 20,000 Hz, but with increasing age the upper limit falls, so older people often cannot hear high-frequency sounds. Hearing loss from loud noise also reduces the upper limit.

11.16. Explain that noise is a nuisance.

Noise is unwanted, unpleasant sound that disturbs work, rest and communication. It causes annoyance, loss of concentration, difficulty in sleeping, and stress. Prolonged exposure to loud noise can damage hearing permanently. For these reasons noise is regarded as a form of pollution and a nuisance in homes, schools, hospitals and workplaces.

11.17. Describe the importance of acoustic protection.

Acoustic protection means reducing unwanted sound by soundproofing walls, using double-glazed windows, fitting soft carpets and curtains, and planting trees around buildings. It is important in hospitals, schools, libraries and recording studios, where quiet is needed, and in factories, where it protects workers' hearing and health.

11.18. What are the uses of ultrasound in medicine?

Ultrasound is used in medicine for imaging internal organs without

surgery, as in scans of a baby before birth. It is used to examine the heart, liver and kidneys, to break kidney stones, to clean teeth, and to measure blood flow. Because it does not use harmful radiation, it is safe for repeated use.

Conceptual Questions

11.1. Why are two tin cans with a string stretched between them a better way to communicate than merely shouting through the air?

The stretched string carries vibrations directly from one can to the other with little loss of energy, whereas sound shouted through air spreads in all directions and loses intensity quickly. The string also guides the vibrations along one path, so the sound arrives clearer and louder at the other end.

11.2. We can recognize persons speaking with the same loudness from their voice. How is this possible?

Every person's voice has a different quality, or timbre, because their vocal cords produce a different mixture of overtones along with the main note. Even when two voices have the same loudness and pitch, the ear recognises the different pattern of overtones, which gives each voice its own character.

11.3. You can listen to your friend round a corner, but you cannot watch him/her. Why?

Sound waves have long wavelengths, comparable to the size of doorways and corners, so they diffract strongly and spread round the corner. Light

waves have extremely short wavelengths, so they hardly diffract at all and travel in straight lines. The sound therefore reaches your ear round the corner, while the light cannot reach your eye.

11.4. Why must the volume of a stereo in a room with wall-to-wall carpet be tuned higher than in a room with a wooden floor?

Carpet is a soft, porous material that absorbs sound energy, so less sound is reflected back into the room. A wooden floor reflects most of the sound, keeping the room louder. To reach the same loudness in the carpeted room, the stereo must therefore be turned up higher to make up for the absorbed energy.

11.5. A student says that the two terms speed and frequency of the wave refer to the same thing. What is your response?

Speed and frequency are different quantities. Speed is how fast the wave disturbance travels through the medium, measured in metres per second. Frequency is how many vibrations pass a point each second, measured in hertz. They are related by speed = frequency x wavelength, but they are not the same thing.

11.6. Two people are listening to the same music at the same distance. They disagree on its loudness. Explain how this could happen.

Loudness is a sensation, not just a physical measurement, and it depends on the listener's ear. The two people may have different hearing sensitivity, for example one may have partial hearing loss. Their ears may also respond differently to different frequencies, so the same intensity level can feel louder to one person than to the other.

11.7. Is there any difference between echo and reflection of sound? Explain.

Reflection of sound is the general phenomenon in which sound bounces back from a surface. An echo is a particular case of reflection in which the reflected sound is heard as a separate, distinct sound after the original sound has ended. An echo is heard only when the reflecting surface is far enough away, about 17 metres or more, for the brain to separate the two sounds.

11.8. Will two separate 50 dB sounds together constitute a 100 dB sound? Explain.

No, two 50 dB sounds together do not make 100 dB. The decibel scale is logarithmic, so adding intensities does not add decibels. Two equal sounds double the intensity, and doubling the intensity raises the level by only about 3 dB, giving about 53 dB in total, not 100 dB.

11.9. Why is ultrasound useful in the medical field?

Ultrasound is useful because it can form images of soft tissues inside the body without cutting and without harmful radiation. Its short wavelength gives fine detail, and it reflects from boundaries between different tissues, allowing doctors to see organs, a developing baby, and blood flow safely and repeatedly.

Numerical Problems

11.1. A normal conversation involves sound intensities of about 3.0 x 10^-6 W/m^2. What is the decibel level for this intensity? What is the

intensity of the sound for 100 dB?

Step 1: Intensity level beta = 10 x log (I / I0), where I0 = 10^-12 W/m^2.

Step 2: I / I0 = (3.0 x 10^-6) / (10^-12) = 3.0 x 10^6. Log of this is 6.48.

Step 3: Beta = 10 x 6.48 = 64.8 dB.

Step 4: For 100 dB: 100 = 10 x log (I / 10^-12), so log (I / 10^-12) = 10, giving I = 10^-12 x 10^10 = 0.01 W/m^2.

Final answer: Decibel level = 64.8 dB; intensity for 100 dB = 0.01 W/m^2.

11.2. If at Anarkali Bazar Lahore, intensity level of sound is 80 dB, what will be the intensity of sound there?

Step 1: Beta = 10 x log (I / I0) = 80 dB, so log (I / I0) = 8.

Step 2: I / I0 = 10^8, with I0 = 10^-12 W/m^2.

Step 3: I = 10^8 x 10^-12 = 10^-4 W/m^2.

Final answer: Intensity = 10^-4 W/m^2.

11.3. At a particular temperature, the speed of sound in air is 330 m/s. If the wavelength of a note is 5 cm, calculate the frequency of the sound wave. Is this frequency in the audible range of the human ear?

Step 1: Wavelength = 5 cm = 0.05 m and speed v = 330 m/s.

Step 2: Frequency f = v / wavelength = 330 / 0.05 = 6600 Hz = 6.6 x 10^3 Hz.

Step 3: The audible range is 20 Hz to 20,000 Hz, and 6600 Hz lies inside it.

Final answer: Frequency = 6.6 x 10^3 Hz; yes, it is audible.

11.4. A doctor counts 72 heartbeats in 1 min. Calculate the frequency and period of the heartbeats.

Step 1: Frequency f = number of beats / time = 72 / 60 s = 1.2 Hz.

Step 2: Period T = 1 / f = 1 / 1.2 = 0.83 s.

Final answer: Frequency = 1.2 Hz, period = 0.83 s.

11.5. A marine survey ship sends a sound wave straight to the seabed. It receives an echo 1.5 s later. The speed of sound in seawater is 1500 m/s. Find the depth of the sea at this position.

Step 1: The sound travels down to the seabed and back, so total distance = speed x time = 1500 x 1.5 = 2250 m.

Step 2: Depth = total distance / 2 = 2250 / 2 = 1125 m.

Final answer: Depth of the sea = 1125 m.

11.6. A student clapped his hands near a cliff and heard the echo after 5 s. What is the distance of the cliff from the student if the speed of the sound is taken as 346 m/s?

Step 1: Total distance travelled = speed x time = 346 x 5 = 1730 m.

Step 2: Distance to the cliff = 1730 / 2 = 865 m.

Final answer: Distance of the cliff = 865 m.

11.7. A ship sends out ultrasound that returns from the seabed and is detected after 3.42 s. If the speed of ultrasound through seawater is 1531 m/s, what is the distance of the seabed from the ship?

Step 1: Total distance = speed x time = 1531 x 3.42 = 5236 m.

Step 2: Distance to the seabed = 5236 / 2 = 2618 m.

Final answer: Distance of the seabed = 2618 m.

11.8. The highest frequency sound humans can hear is about 20,000 Hz. What is the wavelength of sound in air at this frequency at a temperature of 20 degrees C? What is the wavelength of the lowest sounds we can hear of about 20 Hz? Assume the speed of sound in air at 20 degrees C is 343 m/s.

Step 1: For 20,000 Hz: wavelength = speed / frequency = 343 / 20,000 = 0.01715 m, about 1.7 cm.

Step 2: For 20 Hz: wavelength = 343 / 20 = 17.15 m, about 17 m.

Final answer: Wavelength at 20,000 Hz = 1.7 cm; at 20 Hz = 17 m.

Unit 12: Geometrical Optics

Multiple Choice Questions

i. Which of the following quantity is not changed during refraction of light?

(a) its direction

(b) its speed

✓ (c) its frequency

(d) its wavelength

ii. A converging mirror with a radius of 20 cm creates a real image 30 cm from the mirror. What is the object distance?

(a) -5.0 cm

(b) -7.5 cm

✓ (c) -15 cm

(d) -20 cm

Focal length f = R/2 = 20/2 = 10 cm. From 1/f = 1/p + 1/q: 1/p = 1/10 - 1/30 = 1/15, so p = 15 cm (shown as -15 cm in the book's sign convention).

iii. An object is placed at the centre of curvature of a concave mirror. The image produced by the mirror is located

(a) out beyond the centre of curvature

✓ (b) at the centre of curvature

(c) between the centre of curvature and the focal point

(d) at the focal point

iv. An object is 14 cm in front of a convex mirror. The image is 5.8 cm behind the mirror. What is the focal length of the mirror?

(a) -4.1 cm

(b) -8.2 cm

✓ (c) -9.9 cm

(d) -20 cm

1/f = 1/14 - 1/5.8 = 0.0714 - 0.1724 = -0.101, so f = -9.9 cm. The negative sign shows the mirror is convex.

v. The index of refraction depends on

(a) the focal length

✓ (b) the speed of light

(c) the image distance

(d) the object distance

vi. Which type of image is formed by a concave lens?

(a) inverted and real

(b) inverted and virtual

(c) upright and real

✓ (d) upright and virtual

vii. Which type of image is produced by the converging lens of human eye if it views a distant object?

(a) real, erect, same size

✓ (b) real, inverted, diminished

(c) virtual, erect, diminished

(d) virtual, inverted, magnified

viii. Image formed by a camera is

✓ (a) real, inverted, and diminished

(b) virtual, upright and diminished

(c) virtual, upright and magnified

(d) real, inverted and magnified

ix. If a ray of light in glass is incident on an air surface at an angle greater than the critical angle, the ray will

(a) refract only

✓ (b) reflect only

(c) partially refract and partially reflect

(d) diffract only

x. The critical angle for a beam of light passing from water into air is 48.8 degrees. This means that all light rays with an angle of incidence greater than this angle will be

(a) absorbed

✓ (b) totally reflected

(c) partially reflected and partially transmitted

(d) totally transmitted

Review Questions

12.1. What do you understand by reflection of light? Draw a diagram to illustrate reflection at a plane surface.

Reflection of light is the bouncing back of light from a smooth surface into the same medium. In a diagram, draw a straight line for the mirror surface, a normal line at right angles to it, an incident ray striking the surface, and a reflected ray leaving it at the same angle to the normal on the other side.

12.2. Describe the following terms used in reflection: (i) normal (ii) angle of incidence (iii) angle of reflection

The normal is a line drawn at right angles to the reflecting surface at the point where the ray strikes it. The angle of incidence is the angle between the incident ray and the normal. The angle of reflection is the angle between the reflected ray and the normal.

12.3. State laws of reflection. Describe how they can be verified graphically.

The laws of reflection are: the angle of incidence equals the angle of reflection, and the incident ray, the reflected ray and the normal all lie in the same plane. They can be verified by placing pins along the incident and reflected rays on a sheet of paper, joining the pin positions to the point of incidence, and measuring both angles with a protractor to show they are equal.

12.4. Define refraction of light. Describe the passage of light through parallel-sided transparent material.

Refraction is the bending of light when it passes from one transparent medium into another of different optical density, caused by the change in its speed. When light enters a parallel-sided glass slab it bends towards the normal, travels straight inside the slab, and bends away from the normal on leaving, so the emergent ray is parallel to the incident ray but slightly displaced sideways.

12.5. Define the following terms used in refraction: (i) angle of incidence (ii) angle of refraction

The angle of incidence is the angle between the incident ray and the normal to the boundary at the point of incidence. The angle of refraction is the angle between the refracted ray and the normal in the second medium.

12.6. What is meant by refractive index of a material? How would you determine the refractive index of a rectangular glass slab?

The refractive index of a material is the ratio of the speed of light in

vacuum to its speed in that material, and also equals sin(angle of incidence) / sin(angle of refraction). To find it for a glass slab, trace a ray through the slab with pins, measure the angles of incidence and refraction, and divide the sine of the incidence angle by the sine of the refraction angle.

12.7. State the laws of refraction of light and show how they may be verified using rectangular glass slab and pins.

The laws of refraction are: the incident ray, the refracted ray and the normal lie in the same plane, and the ratio sin(i)/sin(r) is constant for a given pair of media (Snell's law). They are verified by tracing rays at several angles of incidence through a glass slab with pins, measuring each angle of refraction, and showing that sin(i)/sin(r) stays the same for all rays.

12.8. What is meant by the term total internal reflection?

Total internal reflection is the complete reflection of light back into a denser medium when it strikes the boundary with a rarer medium at an angle of incidence greater than the critical angle. No light is refracted out, so the boundary acts like a perfect mirror.

12.9. State the conditions for total internal reflection.

Total internal reflection occurs only when two conditions are met: light must travel from a denser medium towards a rarer medium, and the angle of incidence must be greater than the critical angle for that pair of media.

12.10. What is critical angle? Derive a relationship between the critical

angle and the refractive index of a substance.

The critical angle is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90 degrees. From Snell's law, n = sin(i)/sin(r). At the critical angle C, r = 90 degrees and sin(r) = 1, so n = sin(C)/1 is inverted to give sin(C) = 1/n, where n is the refractive index of the denser medium relative to the rarer one.

12.11. What are optical fibres? Describe how total internal reflection is used in light propagating through optical fibres.

Optical fibres are very thin strands of glass or plastic that carry light signals over long distances. Light entering one end strikes the inner wall of the fibre at an angle greater than the critical angle, so it is totally internally reflected again and again along the fibre, following its bends without escaping, until it leaves at the far end.

12.12. Define the following terms applied to a lens: (i) principal axis (ii) optical centre (iii) focal length

The principal axis is the straight line passing through the centres of curvature of the two lens surfaces. The optical centre is the central point of the lens through which a ray passes without bending. The focal length is the distance from the optical centre to the principal focus of the lens.

12.13. What is meant by the principal focus of a (a) convex lens (b) concave lens? Illustrate your answer with ray diagrams.

For a convex lens, the principal focus is the point on the principal axis where rays parallel to the axis meet after refraction. For a concave lens, it is the point from which rays parallel to the axis appear to spread after refraction. In ray diagrams, draw parallel incident rays bending to meet

at the focus for the convex lens, and bending away as if coming from the focus for the concave lens.

12.14. Describe how light is refracted through convex lens.

When a ray passes through a convex lens it bends towards the principal axis. A ray parallel to the axis passes through the principal focus, a ray through the optical centre passes straight through, and a ray through the principal focus leaves parallel to the axis. The combined bending of all rays forms the image.

12.15. With the help of a ray diagram, how you can show the use of thin converging lens as a magnifying glass.

Place a small object between the principal focus and the optical centre of a convex lens. Draw a ray through the centre going straight, and a ray parallel to the axis passing through the focus on the other side. On producing these rays backwards they meet on the same side as the object, forming a virtual, upright and magnified image, which is why the lens works as a magnifying glass.

12.16. A coin is placed at a focal point of a converging lens. Is an image formed? What is its nature?

When the object is at the principal focus, the rays leaving the lens become parallel to each other and never meet. The image is therefore formed at infinity: it is real, inverted and highly magnified, but it cannot be caught on a screen at a finite distance.

12.17. What are the differences between real and virtual images?

A real image is formed by the actual meeting of light rays, can be caught

on a screen, and is usually inverted. A virtual image is formed where rays only appear to meet when produced backwards, cannot be caught on a screen, and is usually upright. Real images are formed by converging lenses and concave mirrors, while virtual images are formed by diverging lenses and convex mirrors.

12.18. How does a converging lens form a virtual image of a real object? How does a diverging lens can form a real image of a real object?

A converging lens forms a virtual image when the object is placed inside its focal length: the refracted rays spread out and only appear to meet behind the lens, giving an upright magnified virtual image. A diverging lens normally forms only virtual images of a real object, but it can form a real image if the incoming light is already converging towards a point beyond the lens, as in a combination of lenses.

12.19. Define power of a lens and its units.

The power of a lens is the reciprocal of its focal length in metres, P = 1/f. It measures how strongly the lens converges or diverges light. Its unit is the dioptre (D): a lens of focal length 1 metre has a power of 1 dioptre, with positive power for convex lenses and negative for concave lenses.

12.20. Describe the passage of light through a glass prism and measure the angle of deviation.

When light enters a glass prism it bends towards the normal, and on leaving it bends away from the normal, so the emergent ray is bent towards the base of the prism. The angle of deviation is the angle between the direction of the incident ray produced forward and the emergent ray. It is measured by tracing both rays on paper and

measuring the angle between them with a protractor.

12.21. Define the terms resolving power and magnifying power.

Resolving power is the ability of an optical instrument to show two close objects as separate rather than merged into one. Magnifying power is the ratio of the angle subtended by the image at the eye to the angle subtended by the object at the eye when viewed at the least distance of distinct vision.

12.22. Draw the ray diagrams of (i) simple microscope (ii) compound microscope (iii) refracting telescope

Simple microscope: one convex lens with the object inside its focus, giving a virtual magnified image. Compound microscope: an objective lens forms a real magnified image inside the focus of the eyepiece, which then magnifies it again into a large virtual image. Refracting telescope: a large objective forms a small real image of a distant object at its focus, and the eyepiece magnifies that image for the eye.

12.23. Mention the magnifying powers of the following optical instruments: (i) simple microscope (ii) compound microscope (iii) refracting telescope

For a simple microscope, magnifying power M = 1 + D/f, where D is the least distance of distinct vision (25 cm). For a compound microscope, M = (L/fo) x (1 + D/fe), where L is the tube length and fo, fe are the focal lengths of objective and eyepiece. For a refracting telescope, M = fo/fe, the ratio of the focal length of the objective to that of the eyepiece.

12.24. Draw ray diagrams to show the formation of images in the normal

human eye.

Draw the eye with cornea and eye lens as a converging system. Parallel rays from a distant object are refracted by the cornea and lens to meet on the retina, forming a real, inverted and diminished image. The brain interprets this image as upright, and the eye lens changes shape to focus nearer objects on the same retina.

12.25. What is meant by the terms nearsightedness and farsightedness? How can these defects be corrected?

Nearsightedness (myopia) is the defect in which distant objects are focused in front of the retina, so distant vision is blurred; it is corrected with a concave (diverging) lens. Farsightedness (hypermetropia) is the defect in which near objects are focused behind the retina, so near vision is blurred; it is corrected with a convex (converging) lens.

Conceptual Questions

12.1. A man raises his left hand in a plane mirror, the image facing him is raising his right hand. Explain why.

A plane mirror reverses front and back, which appears as a left-right reversal. The image's left side faces the man's right side because the mirror turns the man around: the hand that is on the man's left appears on the image's right when the two face each other.

12.2. In your own words, explain why light waves are refracted at a boundary between two materials.

Light travels at different speeds in different materials. When a wavefront

enters a new medium at an angle, one side of the wavefront slows down before the other side, so the whole wavefront turns. This change of direction at the boundary is refraction, and it happens because of the change in speed.

12.3. Explain why a fish under water appears to be at a different depth below the surface than it actually is. Does it appear deeper or shallower?

Light from the fish bends away from the normal when it leaves water into air, so to an observer above the surface the rays appear to come from a higher point. The fish therefore appears shallower than it really is, at about three-quarters of its true depth.

12.4. Why or why not are concave mirrors suitable for makeup?

Concave mirrors are suitable for makeup when the face is placed inside the focal length, because they then give an upright, magnified virtual image. This lets the person see small details clearly. If the face is beyond the focal length, however, the image becomes inverted, so the mirror must be held close to the face.

12.5. Why is the driver's side mirror in many cars convex rather than plane or concave?

A convex mirror always gives an upright, diminished virtual image and has a very wide field of view, so the driver can see much more of the road behind than with a plane mirror. A concave mirror would give a narrow, possibly inverted view and is therefore unsuitable for driving.

12.6. When an optician's testing room is small, he uses a mirror to help him test the eyesight of his patients. Explain why.

Testing eyesight needs a standard viewing distance, usually 6 metres, which a small room cannot provide directly. By placing a mirror on one wall, the optician doubles the light path: the patient reads the chart reflected in the mirror, so the effective distance from eye to chart becomes long enough for the test.

12.7. How does the thickness of a lens affect its focal length?

A thicker lens bends light more strongly because its surfaces are more curved, so it has a shorter focal length and a larger power. A thinner lens of the same material bends light less, giving a longer focal length and smaller power.

12.8. Under what conditions will a converging lens form a virtual image?

A converging lens forms a virtual image when the object is placed between the principal focus and the optical centre, that is, closer to the lens than its focal length. The refracted rays then diverge and only appear to meet behind the lens, giving an upright, magnified virtual image.

12.9. Under what conditions will a converging lens form a real image that is the same size as the object?

A converging lens forms a real image of the same size as the object when the object is placed at twice the focal length (at the centre of curvature). The image then forms at twice the focal length on the other side, is real and inverted, and has the same height as the object.

12.10. Why do we use refracting telescope with large objective lens of large focal length?

A large objective lens collects more light, so faint distant stars appear brighter, and it also gives better resolving power to separate close objects. A large focal length gives a larger image and a higher magnifying power when combined with a short-focus eyepiece.

Numerical Problems

12.1. An object 10.0 cm in front of a convex mirror forms an image 5.0 cm behind the mirror. What is the focal length of the mirror?

Step 1: Use 1/f = 1/p + 1/q, with p = 10 cm and q = -5 cm (image behind the mirror).

Step 2: 1/f = 1/10 - 1/5 = 0.1 - 0.2 = -0.1.

Step 3: f = -10 cm.

Final answer: Focal length = -10 cm (negative sign confirms a convex mirror).

12.2. An object 30 cm tall is located 10.5 cm from a concave mirror with focal length 16 cm. (a) Where is the image located? (b) How high is it?

Step 1: 1/q = 1/f - 1/p = 1/16 - 1/10.5 = 0.0625 - 0.0952 = -0.0327.

Step 2: q = -30.54 cm, so the image is 30.54 cm behind the mirror (virtual).

Step 3: Magnification m = -q/p = 30.54/10.5 = 2.91. Image height = 30 x 2.91 = 87.3 cm.

Final answer: (a) 30.54 cm behind the mirror (b) 87.26 cm.

12.3. An object and its image in a concave mirror are of the same height, yet inverted, when the object is 20 cm from the mirror. What is the focal length of the mirror?

Step 1: An inverted image of the same height forms only when the object is at the centre of curvature.

Step 2: Centre of curvature C = object distance = 20 cm.

Step 3: Focal length f = C/2 = 20/2 = 10 cm.

Final answer: Focal length = 10 cm.

12.4. Find the focal length of a mirror that forms an image 5.66 cm behind the mirror of an object placed at 34.4 cm in front of the mirror. Is the mirror concave or convex?

Step 1: 1/f = 1/34.4 - 1/5.66 = 0.0291 - 0.1767 = -0.1476.

Step 2: f = -6.77 cm.

Step 3: The negative focal length shows the mirror is convex.

Final answer: Focal length = -6.77 cm; convex mirror.

12.5. An image of a statue appears to be 11.5 cm behind a concave mirror with focal length 13.5 cm. Find the distance from the statue to the mirror.

Step 1: Use the mirror formula in the book's form: 1/p = 1/q - 1/f with q = 11.5 cm and f = 13.5 cm.

Step 2: 1/p = 1/11.5 - 1/13.5 = 0.0870 - 0.0741 = 0.0129.

Step 3: p = 77.62 cm.

Final answer: Distance of the statue from the mirror = 77.62 cm.

12.6. An image is produced by a concave mirror of focal length 8.7 cm. The object is 13.2 cm tall and at a distance 19.3 cm from the mirror. (a) Find the location and height of the image. (b) Find the height of the image produced by the mirror if the object is twice as far from the mirror.

Step 1: (a) 1/q = 1/8.7 - 1/19.3 = 0.1149 - 0.0518 = 0.0631, so q = 15.84 cm.

Step 2: Magnification m = q/p = 15.84/19.3 = 0.821. Image height = 13.2 x 0.821 = 10.83 cm.

Step 3: (b) At twice the distance, p = 38.6 cm: 1/q = 1/8.7 - 1/38.6 = 0.0890, q = 11.23 cm, m = 0.291, height = 3.84 cm by the mirror formula (the book prints 5.42 cm for this part, which does not follow from the given data).

Final answer: (a) 15.84 cm, 10.83 cm (b) 3.84 cm by calculation; book answer 5.42 cm flagged.

12.7. Nabeela uses a concave mirror when applying makeup. The mirror has a radius of curvature of 38 cm. (a) What is the focal length of the mirror? (b) Nabeela is located 50 cm from the mirror. Where will her image appear? (c) Will the image be upright or inverted?

Step 1: (a) Focal length f = R/2 = 38/2 = 19 cm.

Step 2: (b) 1/q = 1/19 - 1/50 = 0.0526 - 0.0200 = 0.0326, so q = 30.64 cm in front of the mirror.

Step 3: (c) The object is beyond the centre of curvature, so the image is real and inverted (the book prints 'upright' for this part, which applies only when the face is inside the focal length).

Final answer: (a) 19 cm (b) 30.64 cm (c) inverted (real image).

12.8. An object 4 cm high is placed at a distance of 12 cm from a convex lens of focal length 8 cm. Calculate the position and size of the image. Also state the nature of the image.

Step 1: 1/q = 1/f - 1/p = 1/8 - 1/12 = 0.125 - 0.0833 = 0.0417.

Step 2: q = 24 cm.

Step 3: Magnification m = q/p = 24/12 = 2. Image height = 4 x 2 = 8 cm.

Final answer: Image at 24 cm, 8 cm high, real, inverted and magnified.

12.9. An object 10 cm high is placed at a distance of 20 cm from a concave lens of focal length 15 cm. Calculate the position and size of the image. Also, state the nature of the image.

Step 1: For a concave lens f = -15 cm. 1/q = 1/f - 1/p = -1/15 - 1/20 = -0.0667 - 0.05 = -0.1167.

Step 2: q = -8.57 cm, so the image is 8.57 cm in front of the lens.

Step 3: Magnification m = q/p = 8.57/20 = 0.428. Image height = 10 x 0.428 = 4.28 cm.

Final answer: Image at -8.57 cm, 4.28 cm high, virtual, erect and diminished.

12.10. A convex lens of focal length 6 cm is to be used to form a virtual image three times the size of the object. Where must the lens be placed?

Step 1: Magnification m = 3 for a virtual image, and m = q/p with the lens formula 1/f = 1/p - 1/q (virtual image).

Step 2: With q = 3p (magnitudes): 1/6 = 1/p - 1/(3p) = 2/(3p).

Step 3: 3p = 12, so p = 4 cm.

Final answer: The object must be placed 4 cm from the lens.

12.11. A ray of light from air is incident on a liquid surface at an angle of incidence 35 degrees. Calculate the angle of refraction if the refractive index of the liquid is 1.25. Also calculate the critical angle between the liquid-air interface.

Step 1: Snell's law: sin(r) = sin(i)/n = sin(35)/1.25 = 0.5736/1.25 = 0.4589.

Step 2: Angle of refraction r = 27.31 degrees.

Step 3: Critical angle C: sin(C) = 1/n = 1/1.25 = 0.8, so C = 53.13 degrees.

Final answer: Angle of refraction = 27.31 degrees; critical angle = 53.13 degrees.

12.12. The power of a convex lens is 5 D. At what distance the object should be placed from the lens so that its real and 2 times larger image is formed.

Step 1: Focal length f = 1/P = 1/5 = 0.2 m = 20 cm.

Step 2: Real image, magnification 2, so image distance = 2 x object distance. Lens formula 1/f = 1/v - 1/u with u = -p and v = +2p gives 1/20 = 1/(2p) + 1/p = 3/(2p).

Step 3: 2p = 3 x 20 = 60, so p = 30 cm.

Final answer: Object distance = 30 cm.

Unit 13: Electrostatics

Multiple Choice Questions

i. A positive electric charge

(a) attracts other positive charge

✓ (b) repels other positive charge

(c) attracts a neutral charge

(d) repels a neutral charge

ii. An object gains excess negative charge after being rubbed against another object, which is:

(a) neutral

(b) negatively charged

✓ (c) positively charged

(d) either a, b or c

When an object gains excess negative charge (electrons), the other object loses electrons and becomes positively charged.

iii. Two uncharged objects A and B are rubbed against each other. When object B is placed near a negatively charged object C, the two objects repel each other. Which of the following statements is true about object A?

(a) remains uncharged

✓ (b) becomes positively charged

(c) becomes negatively charged

(d) unpredictable

B repels the negative object C, so B is also negative. Rubbing transfers charge, so A lost electrons to B and became positive.

iv. When you rub a plastic rod against your hair several times and put it near some bits of paper, the pieces of paper are attracted towards it. What does this observation indicate?

(a) the rod and the paper are oppositely charged

(b) the rod acquires a positive charge

(c) the rod and the paper have the same charges

✓ (d) the rod acquires a negative charge

Rubbing plastic against hair gives the rod excess electrons, so it becomes negatively charged and attracts the neutral paper by induction.

v. According to Coulomb's law, what happens to the attraction of two oppositely charged objects as their distance of separation increases?

(a) increases

✓ (b) decreases

(c) remains unchanged

(d) cannot be determined

vi. Coulomb's law is valid for the charges which are

(a) moving and point charges

(b) moving and non-point charges

✓ (c) stationary and point charges

(d) stationary and large size charges

vii. A positive and a negative charge are initially 4 cm apart. When they

are moved closer together so that they are now only 1 cm apart, the force between them is

(a) 4 times smaller than before

(b) 4 times larger than before

(c) 8 times larger than before

✓ (d) 16 times larger than before

Force is inversely proportional to distance squared. Distance becomes 1/4, so force becomes 16 times larger.

viii. Five joules of work is needed to shift 10 C of charge from one place to another. The potential difference between the places is

✓ (a) 0.5 V

(b) 2 V

(c) 5 V

(d) 10 V

Potential difference = work / charge = 5 / 10 = 0.5 V.

ix. Two small charged spheres are separated by 2 mm. Which of the following would produce the greatest attractive force?

(a) +1q and +4q

(b) -1q and -4q

✓ (c) +2q and +2q

(d) +2q and -2q

Attraction needs opposite signs. Among the choices, the opposite-sign pair is +2q and -2q, and its product of magnitudes (4q^2) equals the largest product, giving the greatest attractive force at the same distance.

x. Electric field lines

(a) always cross each other

✓ (b) never cross each other

(c) cross each other in the region of strong field

(d) cross each other in the region of weak field

xi. Capacitance is defined as

(a) V x C

✓ (b) Q / V

(c) Q x V

(d) V / Q

Review Questions

13.1. How can you show by simple experiments that there are two types of electric charges?

Rub a glass rod with silk and suspend it freely; it repels another glass rod rubbed with silk but attracts an ebonite rod rubbed with fur. This shows there are two kinds of charge: charges that behave like the glass rod are called positive, and those that behave like the ebonite rod are called negative. Like charges repel and unlike charges attract.

13.2. Describe the method of charging bodies by electrostatic induction.

Bring a charged rod near a neutral conductor without touching it. The free electrons in the conductor are repelled to the far end, leaving the near end positively charged. While the rod is still near, earth the far end so electrons escape, then remove the earth connection and then the rod. The conductor is left with a net positive charge, opposite to the rod's

charge.

13.3. How does electrostatic induction differ from charging by friction?

In charging by friction, two bodies are rubbed together and electrons are actually transferred from one body to the other, so both bodies become oppositely charged. In electrostatic induction, the charging body never touches the conductor; it only separates the charges already in the conductor, and earthing removes one kind of charge, leaving a net charge without contact.

13.4. What is gold leaf electroscope? Discuss its working principle with a labelled diagram.

A gold leaf electroscope is an instrument for detecting and testing electric charge. It has a metal rod passing through an insulating stopper into a glass case, with a metal cap on top and two thin gold leaves hanging from the lower end of the rod. When a charged body touches the cap, charge spreads to the leaves, which repel each other and diverge because they carry the same charge. The amount of divergence shows the amount of charge.

13.5. Suppose you have a glass rod which becomes positively charged when you rub it with wool. Describe how would you charge the electroscope (i) negatively (ii) positively.

(i) To charge negatively by induction: bring the positive glass rod near the cap, earth the cap briefly, remove the earth and then the rod; the electroscope is left negatively charged. (ii) To charge positively: touch the cap directly with the positively charged glass rod, so positive charge flows onto the electroscope and the leaves diverge.

13.6. With the help of electroscope how you can find presence of charge on a body.

Bring the body near the cap of an uncharged electroscope without touching it. If the body is charged, it will induce a separation of charge in the electroscope and the gold leaves will diverge. If the body is uncharged, the leaves will not move at all, showing that no charge is present on the body.

13.7. Describe how you would determine the nature of the charge on a body by using electroscope.

First charge the electroscope with a known charge, for example positively, so its leaves are diverged. Then bring the unknown body near the cap: if the leaves diverge further, the body has the same (positive) charge; if the leaves collapse, the body has the opposite (negative) charge. Attraction alone is not enough, because a neutral body can also attract by induction, so the increase or decrease in divergence decides the nature.

13.8. Explain Coulomb's law of electrostatics and write its mathematical form.

Coulomb's law states that the force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them. Its mathematical form is F = k x q1 x q2 / r^2, where k = 9 x 10^9 N m^2/C^2 in vacuum. The force is attractive for unlike charges and repulsive for like charges, acting along the line joining them.

13.9. What is meant by electric field and electric intensity?

The electric field is the region around a charged body in which another charge experiences a force. Electric intensity at a point is the force acting on a unit positive charge placed at that point, E = F/q. Its unit is newton per coulomb (N/C), and it is measured by placing a small test charge at the point.

13.10. Is electric intensity a vector quantity? What will be its direction?

Yes, electric intensity is a vector quantity because force is a vector. Its direction is the direction of the force on a positive test charge placed at that point: away from a positive charge and towards a negative charge. Field lines are drawn in this direction.

13.11. How would you define potential difference between two points? Define its unit.

The potential difference between two points is the work done in moving a unit positive charge from one point to the other, V = W/q. Its unit is the volt (V): one volt is the potential difference when one joule of work is done in moving one coulomb of charge between the points.

13.12. Show that potential difference can be described as energy transfer per unit charge between the two points.

When a charge q moves through a potential difference V, the work done on it is W = qV, so V = W/q. This means each coulomb of charge gains or loses V joules of energy in moving between the points. Potential difference is therefore the energy transferred per unit charge, measured in joules per coulomb, which is the volt.

13.13. What do you mean by the capacitance of a capacitor? Define units

of capacitance.

Capacitance is the ability of a capacitor to store charge per unit potential difference, C = Q/V. It depends on the area of the plates, their separation, and the medium between them. Its unit is the farad (F): a capacitor has a capacitance of one farad if it stores one coulomb of charge when the potential difference across it is one volt.

13.14. Derive the formula for the equivalent capacitance for a series combination of a number of capacitors.

In series, each capacitor carries the same charge Q, and the total voltage is the sum of the voltages across each: V = V1 + V2 + V3. Since V = Q/C, this gives Q/Ceq = Q/C1 + Q/C2 + Q/C3. Cancelling Q gives 1/Ceq = 1/C1 + 1/C2 + 1/C3, so the equivalent capacitance is smaller than any single capacitance in the series.

13.15. Discuss different types of capacitors.

Common types of capacitors are paper capacitors, in which paper soaked in wax separates metal foils; mica capacitors, which use mica sheets and are very stable; ceramic capacitors, which are small and used in radio circuits; electrolytic capacitors, which give large capacitance in a small size but must be connected with correct polarity; and variable capacitors, whose capacitance can be changed by moving the plates, used for tuning radios.

13.16. What is difference between variable and fixed type capacitor?

A fixed capacitor has a constant capacitance that cannot be changed, and it is used where a steady value is needed in a circuit. A variable capacitor has plates whose overlap can be adjusted, so its capacitance

can be varied over a range; it is used for tuning, for example to select a radio station.

13.17. Enlist some uses of capacitors.

Capacitors are used to store electric charge and energy, to smooth the output of power supplies, to block direct current while passing alternating current, to tune radio and television circuits, to start electric motors, in camera flashes to release a sudden burst of energy, and in computer memory circuits.

13.18. Discuss one application of static electricity.

One important application of static electricity is the electrostatic precipitator used in factories and power stations. Smoke particles are given an electric charge as they pass through a charged grid, and are then attracted to oppositely charged plates, so the smoke is cleaned before it leaves the chimney and air pollution is reduced.

13.19. What are hazards of static electricity?

Static electricity can cause sparks that ignite petrol vapour, gas or dust in factories and fuel stations, leading to fires and explosions. It can give painful shocks, damage delicate electronic components, and attract dust onto surfaces. Lightning, the largest form of static discharge, can damage buildings, start fires and kill people.

Conceptual Questions

13.1. An electrified rod attracts pieces of paper. After a while these pieces fly away! Why?

At first the charged rod attracts the neutral paper by induction, because the near side of each piece acquires opposite charge. After contact, each piece gains the same charge as the rod, so it is then repelled and flies away. The pieces therefore first stick to the rod and then jump off it.

13.2. How much negative charge has been removed from a positively charged electroscope, if it has a charge of 7.5 x 10^-11 C?

A positively charged electroscope has a deficiency of electrons, so its positive charge equals the charge of the electrons removed. Removing electrons of total charge 7.5 x 10^-11 C leaves the electroscope with a net positive charge of 7.5 x 10^-11 C, since each electron carries 1.6 x 10^-19 C and about 4.7 x 10^8 electrons have been removed.

13.3. In what direction will a positively charged particle move in an electric field?

A positively charged particle moves in the direction of the electric field, that is, along the field lines away from a positive charge and towards a negative charge. This is because the electric force on a positive charge acts in the same direction as the field intensity.

13.4. Does each capacitor carry equal charge in series combination? Explain.

Yes, in a series combination each capacitor carries the same charge. When the combination is connected to a battery, the battery moves

charge onto the first plate, which induces an equal and opposite charge on the second plate, and this induction continues through the whole chain. Every capacitor therefore ends up with the same magnitude of charge, although their voltages differ.

13.5. Each capacitor in parallel combination has equal potential difference between its two plates. Justify the statement.

In a parallel combination, one plate of every capacitor is connected directly to one terminal of the battery and the other plate to the other terminal. All the capacitors are therefore connected across the same two points, so each one experiences the full battery voltage. Hence the potential difference across each capacitor is the same, while their charges differ according to their capacitances.

13.6. Perhaps you have seen a gasoline truck trailing a metal chain beneath it. What purpose does the chain serve?

As a gasoline truck moves, friction can build up static charge on its body. The trailing metal chain touches the ground continuously and conducts this charge safely to the earth, preventing a spark. Without the chain, a spark could ignite the petrol vapour and cause a fire or explosion.

13.7. If a high-voltage power line fell across your car while you were in the car, why should you not come out of the car?

The metal body of the car acts like a cage: the charge stays on the outside of the car and the inside remains safe. If you step out while touching the car and the ground at the same time, the current can pass through your body to the earth and electrocute you. You should stay inside until the power is switched off and help arrives.

13.8. Explain why a glass rod can be charged by rubbing when held by hand but an iron rod cannot be charged by rubbing, if held by hand?

Glass is an insulator, so the charge produced by rubbing stays where it is formed on the rod. Iron is a conductor, so any charge produced flows at once through the hand and body to the earth, leaving the rod uncharged. An iron rod can be charged by rubbing only if it is held by an insulating handle.

Numerical Problems

13.1. The charge of how many negatively charged particles would be equal to 100 microC? Assume charge on one negative particle is 1.6 x 10^-19 C.

Step 1: Total charge Q = 100 microC = 100 x 10^-6 C = 10^-4 C.

Step 2: Number of particles n = Q / e = 10^-4 / (1.6 x 10^-19).

Step 3: n = 6.25 x 10^14.

Final answer: 6.25 x 10^14 particles.

13.2. Two point charges q1 = 10 microC and q2 = 5 microC are placed at a distance of 150 cm. What will be the Coulomb's force between them? Also find the direction of the force.

Step 1: r = 150 cm = 1.5 m; q1 = 10 x 10^-6 C; q2 = 5 x 10^-6 C.

Step 2: F = k x q1 x q2 / r^2 = (9 x 10^9) x (10 x 10^-6) x (5 x 10^-6) / (1.5)^2.

Step 3: F = (9 x 10^9 x 50 x 10^-12) / 2.25 = 0.45 / 2.25 = 0.2 N.

Step 4: Both charges are positive, so the force is repulsive, acting away

from each other.

Final answer: 0.2 N, repulsion.

13.3. The force of repulsion between two identical positive charges is 0.8 N, when the charges are 0.1 m apart. Find the value of each charge.

Step 1: F = k x q^2 / r^2, so q^2 = F x r^2 / k.

Step 2: q^2 = 0.8 x (0.1)^2 / (9 x 10^9) = 0.008 / (9 x 10^9) = 8.89 x 10^-13.

Step 3: q = square root (8.89 x 10^-13) = 9.4 x 10^-7 C.

Final answer: Each charge = 9.4 x 10^-7 C.

13.4. Two charges repel each other with a force of 0.1 N when they are 5 cm apart. Find the force between the same charges when they are 2 cm apart.

Step 1: Force is inversely proportional to distance squared: F2 / F1 = (r1 / r2)^2.

Step 2: F2 = 0.1 x (5/2)^2 = 0.1 x 6.25 = 0.625 N, about 0.62 N.

Final answer: Force = 0.62 N.

13.5. The electric potential at a point in an electric field is 10^4 V. If a charge of +100 microC is brought from infinity to this point, what would be the amount of work done on it?

Step 1: Work W = qV.

Step 2: q = 100 microC = 100 x 10^-6 C = 10^-4 C; V = 10^4 V.

Step 3: W = 10^-4 x 10^4 = 1 J.

Final answer: Work done = 1 J.

13.6. A point charge of +2 C is transferred from a point at potential 100 V to a point at potential 50 V. What would be the energy supplied by the charge?

Step 1: Potential difference = 100 - 50 = 50 V.

Step 2: Energy supplied = q x potential difference = 2 x 50 = 100 J.

Final answer: Energy supplied = 100 J.

13.7. A capacitor holds 0.06 coulombs of charge when fully charged by a 9 volt battery. Calculate capacitance of the capacitor.

Step 1: C = Q / V = 0.06 / 9.

Step 2: C = 0.00667 F = 6.67 x 10^-3 F.

Final answer: Capacitance = 6.67 x 10^-3 F.

13.8. A capacitor holds 0.03 coulombs of charge when fully charged by a 6 volt battery. How much voltage would be required for it to hold 2 coulombs of charge?

Step 1: Capacitance C = Q / V = 0.03 / 6 = 0.005 F.

Step 2: For Q = 2 C: V = Q / C = 2 / 0.005 = 400 V.

Final answer: Voltage required = 400 V.

13.9. Two capacitors of capacitances 6 microF and 12 microF are connected in series with 12 V battery. Find the equivalent capacitance of

the combination. Find the charge and the potential difference across each capacitor.

Step 1: 1/Ceq = 1/6 + 1/12 = 3/12 = 1/4, so Ceq = 4 microF.

Step 2: Charge (same for both) Q = Ceq x V = 4 x 12 = 48 microC.

Step 3: Voltage across 6 microF = 48/6 = 8 V; across 12 microF = 48/12 = 4 V.

Final answer: Ceq = 4 microF, charge = 48 microC, voltages 8 V and 4 V.

13.10. Two capacitors of capacitances 6 microF and 12 microF are connected in parallel with a 12 V battery. Find the equivalent capacitance of the combination. Find the charge and the potential difference across each capacitor.

Step 1: Ceq = 6 + 12 = 18 microF.

Step 2: Voltage across each = 12 V (parallel).

Step 3: Charge on 6 microF = 6 x 12 = 72 microC; on 12 microF = 12 x 12 = 144 microC.

Final answer: Ceq = 18 microF, charges 72 microC and 144 microC, voltage 12 V each.

Unit 14: Current Electricity

Multiple Choice Questions

i. An electric current in conductors is due to the flow of

(a) positive ions

(b) negative ions

(c) positive charges

✓ (d) free electrons

ii. What is the voltage across a 6 ohm resistor when 3 A of current passes through it?

(a) 2 V

(b) 9 V

✓ (c) 18 V

(d) 36 V

V = I x R = 3 x 6 = 18 V.

iii. What happens to the intensity or the brightness of the lamps connected in series as more and more lamps are added?

(a) increases

✓ (b) decreases

(c) remains the same

(d) cannot be predicted

iv. Why should household appliances be connected in parallel with the voltage source?

(a) to increase the resistance of the circuit

(b) to decrease the resistance of the circuit

✓ (c) to provide each appliance the same voltage as the power source

(d) to provide each appliance the same current as the power source

v. Electric potential and e.m.f.

(a) are the same terms

✓ (b) are the different terms

(c) have different units

(d) both (b) and (c)

vi. When we double the voltage in a simple electric circuit, we double the

✓ (a) current

(b) power

(c) resistance

(d) both (a) and (b)

By Ohm's law, current is proportional to voltage at fixed resistance, so doubling voltage doubles current. Power, which is V x I, would actually become four times, so the book's intended answer is current.

vii. If we double both the current and the voltage in a circuit while keeping its resistance constant, the power

(a) remains unchanged

(b) halves

(c) doubles

✓ (d) quadruples

Power P = V x I. Doubling both V and I gives P = 2V x 2I = 4VI, so power becomes four times.

viii. What is the power rating of a lamp connected to a 12 V source when it carries 2.5 A?

(a) 4.8 W

(b) 14.5 W

✓ (c) 30 W

(d) 60 W

P = V x I = 12 x 2.5 = 30 W.

ix. The combined resistance of two identical resistors, connected in series is 8 ohm. Their combined resistance in a parallel arrangement will be

✓ (a) 2 ohm

(b) 4 ohm

(c) 8 ohm

(d) 12 ohm

Each resistor is 4 ohm (series: R + R = 8). In parallel: 1/Req = 1/4 + 1/4 = 1/2, so Req = 2 ohm.

Review Questions

14.1. Define and explain the term electric current.

Electric current is the rate of flow of electric charge through a conductor, I = Q/t. In a metal wire it is the flow of free electrons, and it is measured in amperes: one ampere is one coulomb of charge passing a point in one second.

14.2. What is the difference between electronic current and conventional current?

Electronic current is the actual flow of electrons, from the negative terminal to the positive terminal of a source. Conventional current is the

direction in which positive charge would flow, from positive to negative terminal, and it is the direction used in circuit diagrams. The two directions are opposite to each other.

14.3. What do we mean by the term e.m.f? Is it really a force? Explain.

E.m.f. (electromotive force) is the energy supplied by a source per unit charge, measured in volts. It is not really a force: the name is historical. A cell of e.m.f. 1.5 V supplies 1.5 joules of energy to each coulomb of charge passing through it.

14.4. How can we differentiate between e.m.f. and potential difference?

E.m.f. is the total energy supplied per unit charge by a source when no current flows, or the work done by the source. Potential difference is the energy used per unit charge between two points of a circuit when current flows. E.m.f. is measured across the source on open circuit, while potential difference is measured across a component in a working circuit and is less than the e.m.f. because some energy is lost in the internal resistance.

14.5. Explain Ohm's law. What are its limitations?

Ohm's law states that the current through a conductor is directly proportional to the potential difference across it, provided its temperature and physical state remain constant: V = IR. Its limitations are that it holds only for ohmic conductors at constant temperature; it fails for diodes, transistors, filament lamps and electrolytes, whose resistance changes with current or voltage.

14.6. Define resistance and its units.

Resistance is the opposition offered by a conductor to the flow of current, R = V/I. Its unit is the ohm (symbol ohm): a conductor has a resistance of one ohm if a potential difference of one volt across it produces a current of one ampere.

14.7. What is the difference between conductors and insulators?

Conductors are materials that allow electric charge to flow easily because they have many free electrons, for example copper, aluminium and iron. Insulators are materials that do not allow charge to flow easily because their electrons are tightly bound, for example rubber, plastic, wood and glass.

14.8. Explain the energy dissipation in a resistance. What is Joule's law?

When current flows through a resistance, the moving electrons collide with the atoms of the conductor and give up energy, which appears as heat. Joule's law states that the heat produced is H = I^2 x R x t, where I is current, R is resistance and t is time. This heating effect is used in heaters and bulbs but is a loss in power transmission.

14.9. What is difference between D.C and A.C?

Direct current (D.C.) flows always in one direction with a steady value, as supplied by cells and batteries. Alternating current (A.C.) repeatedly reverses direction and changes in magnitude, as supplied by generators and the mains. Household supply in Pakistan is 220 V A.C. at 50 Hz.

14.10. Discuss the main features of parallel combination of resistors.

In a parallel combination, each resistor is connected across the same voltage, so the voltage across each is equal. The total current divides

among the branches in inverse proportion to their resistances. The equivalent resistance is given by 1/Req = 1/R1 + 1/R2 + ..., and it is smaller than the smallest single resistance in the combination.

14.11. Determine the equivalent resistance of series combination of resistors.

In series, the same current flows through every resistor and the total voltage is the sum of the voltages across each: V = V1 + V2 + V3. Since V = IR, this gives I x Req = I x R1 + I x R2 + I x R3. Cancelling I gives Req = R1 + R2 + R3, so the equivalent resistance is the sum of the separate resistances.

14.12. Describe briefly the hazards of household electricity.

Household electricity can cause electric shocks that may be fatal, fires from short circuits and overloaded wires, and burns from contact with live parts. Damaged insulation, wet hands, and faulty appliances increase the danger, and a short circuit can melt wires and set fire to nearby materials.

14.13. Describe four safety measures that should be taken in connection with the household circuit.

Four safety measures are: use fuses or circuit breakers that break the circuit on excess current; earth all metal appliances so a fault current flows safely to the ground; use insulated wires and switches and replace damaged ones at once; and never touch switches or appliances with wet hands, and keep water away from sockets.

14.14. Design a circuit diagram for a study room that needs the following

equipments in parallel: (a) One 100 W lamp operated by one switch. (b) One reading lamp fitted with a 40 W bulb which can be switched ON and OFF from two points. (c) What is the advantage of connecting the equipments in parallel instead of series?

Draw the mains supply with a live and a neutral line. Connect the 100 W lamp in one branch with its own switch in series with it. Connect the 40 W reading lamp in a second branch with two two-way switches, one at each end of the room, wired so either switch can turn it on or off. (c) In parallel, each appliance gets the full mains voltage and can be switched independently, and if one fails the others keep working; in series they would share the voltage and all would go off together.

Conceptual Questions

14.1. Why in conductors is charge transferred by free electrons rather than by positive charges?

In a metal conductor the positive ions are fixed in the crystal lattice and cannot move, while a large number of electrons are free to move through the lattice. When a voltage is applied, these free electrons drift towards the positive terminal, so charge is carried by electrons and not by the fixed positive ions.

14.2. What is the difference between a cell and a battery?

A cell is a single unit that converts chemical energy into electrical energy, for example a 1.5 V dry cell. A battery is a combination of two or more cells connected together, usually in series, to give a larger voltage, for example a 12 V car battery made of six 2 V cells.

14.3. Can current flow in a circuit without potential difference?

No, a steady current cannot flow without a potential difference. Current is the flow of charge driven by an electric field, which exists only when there is a potential difference between two points. Without it, the free electrons move randomly but there is no net flow in any direction.

14.4. Two points on an object are at different electric potentials. Does charge necessarily flow between them?

Not necessarily. Charge flows only if there is a conducting path between the points. If the points are separated by an insulator, no charge can flow even though a potential difference exists, just as a charged capacitor holds its charge until a conducting path is provided.

14.5. In order to measure current in a circuit why is an ammeter always connected in series?

An ammeter measures the current passing through it, so it must be placed in the path of the current, that is, in series with the component. It has a very low resistance so it does not change the circuit current appreciably. Connected in parallel it would short-circuit the component and be damaged by the large current.

14.6. In order to measure voltage in a circuit why is a voltmeter always connected in parallel? Discuss.

A voltmeter measures the potential difference between two points, so it must be connected across the component, that is, in parallel with it. It has a very high resistance so it draws almost no current and does not disturb the circuit. Connected in series it would block the current and give a wrong reading.

14.7. How many watt-hours are there in 1000 joules?

One watt-hour is the energy used by one watt for one hour: 1 Wh = 1 W x 3600 s = 3600 J. Therefore 1000 J = 1000/3600 Wh = 0.278 Wh, about 0.28 watt-hours.

14.8. From your experience in watching cars on the roads at night, are automobile headlamps connected in series or in parallel?

Headlamps are connected in parallel. This is clear because when one headlamp fails, the other keeps shining, and both shine with full brightness. In series, one failure would switch off both lamps and each would get only half the voltage.

14.9. A certain flash-light can use a 10 ohm bulb or a 5 ohm bulb. Which bulb should be used to get the brighter light? Which bulb will discharge the battery first?

The 5 ohm bulb should be used for brighter light, because at the same battery voltage it draws more current (I = V/R) and uses more power (P = V^2/R). For the same reason, the 5 ohm bulb will discharge the battery first, since it consumes energy faster.

14.10. It is impracticable to connect an electric bulb and an electric heater in series. Why?

A bulb and a heater need very different currents and voltages: the heater draws a large current while the bulb needs a small one. In series the same current must flow through both, so neither would work at its rated value, the bulb would be dim or could burn out, and switching off one would switch off the other.

14.11. Does a fuse in a circuit control the potential difference or the current?

A fuse controls the current. It is a thin wire that melts and breaks the circuit when the current exceeds its safe rating, protecting the wiring and appliances. It does not control the potential difference, which is fixed by the supply.

Numerical Problems

14.1. A current of 3 mA is flowing through a wire for 1 minute. What is the charge flowing through the wire?

Step 1: Q = I x t. I = 3 mA = 3 x 10^-3 A; t = 1 minute = 60 s.

Step 2: Q = 3 x 10^-3 x 60 = 180 x 10^-3 C = 0.18 C.

Final answer: Charge = 180 x 10^-3 C.

14.2. At 100,000 ohm, how much current flows through your body if you touch the terminals of a 12 V battery? If your skin is wet, so that your resistance is only 1000 ohm, how much current would you receive from the same battery?

Step 1: I = V/R = 12 / 100,000 = 1.2 x 10^-4 A (dry skin).

Step 2: I = 12 / 1000 = 1.2 x 10^-2 A (wet skin).

Final answer: 1.2 x 10^-4 A dry; 1.2 x 10^-2 A wet.

14.3. The resistance of a conductor wire is 10 megaohm. If a potential difference of 100 volts is applied across its ends, then find the value of

current passing through it in mA.

Step 1: R = 10 megaohm = 10 x 10^6 ohm; V = 100 V.

Step 2: I = V/R = 100 / (10 x 10^6) = 10 x 10^-6 A = 0.01 mA.

Final answer: Current = 0.01 mA.

14.4. By applying a potential difference of 10 V across a conductor, a current of 1.5 A passes through it. How much energy would be obtained from the current in 2 minutes?

Step 1: Energy E = V x I x t. t = 2 minutes = 120 s.

Step 2: E = 10 x 1.5 x 120 = 1800 J.

Final answer: Energy = 1800 J.

14.5. Two resistances of 2 kilo-ohm and 8 kilo-ohm are joined in series, if a 10 V battery is connected across the ends of this combination, find: (a) The equivalent resistance of the series combination. (b) Current passing through each of the resistances. (c) The potential difference across each resistance.

Step 1: (a) Req = 2 + 8 = 10 kilo-ohm.

Step 2: (b) Current I = V/Req = 10 / 10,000 = 0.001 A = 1 mA, the same through both.

Step 3: (c) V across 2 kilo-ohm = 0.001 x 2000 = 2 V; across 8 kilo-ohm = 0.001 x 8000 = 8 V.

Final answer: (a) 10 kilo-ohm (b) 1 mA (c) 2 V, 8 V.

14.6. Two resistances of 6 kilo-ohm and 12 kilo-ohm are connected in

parallel. A 6 V battery is connected across its ends, find: (a) Equivalent resistance of the parallel combination. (b) Current passing through each of the resistances. (c) Potential difference across each of the resistance.

Step 1: (a) 1/Req = 1/6 + 1/12 = 3/12 = 1/4, so Req = 4 kilo-ohm.

Step 2: (b) Current through 6 kilo-ohm = 6/6000 = 1 mA; through 12 kilo-ohm = 6/12000 = 0.5 mA.

Step 3: (c) In parallel, each resistor gets the full battery voltage = 6 V.

Final answer: (a) 4 kilo-ohm (b) 1 mA, 0.5 mA (c) 6 V.

14.7. An electric bulb is marked with 220 V, 100 W. Find the resistance of the filament of the bulb. If the bulb is used 5 hours daily, find the energy in kilowatt-hour consumed by the bulb in one month (30 days).

Step 1: Resistance R = V^2 / P = (220)^2 / 100 = 48,400 / 100 = 484 ohm.

Step 2: Energy per day = 100 W x 5 h = 500 Wh = 0.5 kWh.

Step 3: Energy in 30 days = 0.5 x 30 = 15 kWh.

Final answer: Resistance = 484 ohm; energy = 15 kWh.

14.8. An incandescent light bulb with an operating resistance of 95 ohm is labelled '150 W.' Is this bulb designed for use in a 120 V circuit or a 220 V circuit?

Step 1: Power P = V^2 / R, so V = square root (P x R) = square root (150 x 95) = square root (14,250).

Step 2: V = 119.4 V, about 120 V.

Final answer: The bulb is designed for a 120 V circuit.

14.9. A house is installed with (a) 10 bulbs of 60 W each used 5 hours daily. (b) 4 fans of 75 W each run 10 hours daily. (c) One T.V. of 100 W used 5 hours daily. (d) One electric iron of 1000 W used 2 hours daily. If the cost of one unit of electricity is Rs. 4, find the monthly expenditure of electricity (one month = 30 days).

Step 1: Bulbs: 10 x 60 x 5 = 3000 Wh/day. Fans: 4 x 75 x 10 = 3000 Wh/day.

Step 2: T.V.: 100 x 5 = 500 Wh/day. Iron: 1000 x 2 = 2000 Wh/day.

Step 3: Total per day = 3000 + 3000 + 500 + 2000 = 8500 Wh = 8.5 kWh (units).

Step 4: Monthly units = 8.5 x 30 = 255 units. Cost = 255 x 4 = Rs. 1020.

Final answer: Monthly expenditure = Rs. 1020.

14.10. A 100 W lamp bulb and a 4 kW water heater are connected to a 250 V supply. Calculate (a) the current which flows in each appliance (b) the resistance of each appliance when in use.

Step 1: (a) Lamp current = P/V = 100/250 = 0.4 A. Heater current = 4000/250 = 16 A.

Step 2: (b) Lamp resistance = V/I = 250/0.4 = 625 ohm. Heater resistance = 250/16 = 15.62 ohm.

Final answer: (a) 0.4 A, 16 A (b) 625 ohm, 15.62 ohm.

14.11. A resistor of resistance 5.6 ohm is connected across a battery of 3.0 V by means of a wire of negligible resistance. A current of 0.5 A passes through the resistor. Calculate (a) Power dissipated in the resistor.

(b) Total power produced by the battery. (c) Give the reason of difference between these two quantities.

Step 1: (a) Power in resistor = I^2 x R = (0.5)^2 x 5.6 = 0.25 x 5.6 = 1.4 W.

Step 2: (b) Total power = V x I = 3.0 x 0.5 = 1.5 W.

Step 3: (c) The difference of 0.1 W is lost as heat in the internal resistance of the battery.

Final answer: (a) 1.4 W (b) 1.5 W (c) power lost in internal resistance.

Unit 15: Electromagnetism

Multiple Choice Questions

i. Which statement is true about the magnetic poles?

(a) unlike poles repel

(b) like poles attract

(c) magnetic poles do not affect each other

✓ (d) a single magnetic pole does not exist

ii. What is the direction of the magnetic field lines inside a bar magnet?

(a) from north pole to south pole

✓ (b) from south pole to north pole

(c) from side to side

(d) there are no magnetic field lines

iii. The presence of a magnetic field can be detected by a

(a) small mass

(b) stationary positive charge

(c) stationary negative charge

✓ (d) magnetic compass

iv. If the current in a wire which is placed perpendicular to a magnetic field increases, the force on the wire

✓ (a) increases

(b) decreases

(c) remains the same

(d) will be zero

v. A D.C motor converts

(a) mechanical energy into electrical energy

(b) mechanical energy into chemical energy

✓ (c) electrical energy into mechanical energy

(d) electrical energy into chemical energy

vi. Which part of a D.C motor reverses the direction of current through the coil every half-cycle?

(a) the armature

✓ (b) the commutator

(c) the brushes

(d) the slip rings

vii. The direction of induced e.m.f. in a circuit is in accordance with conservation of

(a) mass

(b) charge

(c) momentum

✓ (d) energy

viii. The step-up transformer

(a) increases the input current

✓ (b) increases the input voltage

(c) has more turns in the primary

(d) has less turns in the secondary coil

ix. The turn ratio of a transformer is 10. It means

(a) Is = 10 Ip

(b) Ns = Np / 10

✓ (c) Ns = 10 Np

(d) Vs = Vp / 10

Review Questions

15.1. Demonstrate by an experiment that a magnetic field is produced around a straight current-carrying conductor.

Pass a straight wire vertically through a horizontal card and sprinkle iron filings on the card. With no current, the filings lie randomly. When current flows through the wire, the filings arrange themselves in concentric circles around the wire, and a compass needle placed near the wire deflects. This shows a magnetic field exists around the current-carrying conductor.

15.2. State and explain the rule by which the direction of the lines of force of the magnetic field around a current-carrying conductor can be determined.

The right-hand grip rule gives the direction: grip the conductor with the right hand so that the thumb points in the direction of the current; the curled fingers then point in the direction of the magnetic field lines around the conductor. Reversing the current reverses the field direction.

15.3. You are given an unmarked magnetized steel bar and a bar magnet, its north and south ends are marked N and S respectively. State how you would determine the polarity at each end of the unmarked bar.

Bring one end of the unmarked bar near the N end of the marked magnet. If it is repelled, that end is also north, because like poles repel; if it is attracted, that end is south. Test the other end the same way to confirm: the end attracted to N is south and the end repelled by N is north.

15.4. When a straight current-carrying conductor is placed in a magnetic field, it experiences a force. State the rule by which the direction of this force can be found out.

Fleming's left-hand rule gives the direction: hold the thumb, first finger and second finger of the left hand mutually at right angles. The first finger points in the direction of the magnetic field, the second finger in the direction of the current, and the thumb then points in the direction of the force on the conductor.

15.5. State that a current-carrying coil in a magnetic field experiences a torque.

When a rectangular coil carrying current is placed in a magnetic field, the two opposite sides of the coil experience equal and opposite forces in different directions, forming a couple. This couple produces a torque that rotates the coil. The torque is greatest when the plane of the coil is parallel to the field and zero when it is perpendicular to it.

15.6. What is an electric motor? Explain the working principle of D.C motor.

An electric motor is a device that converts electrical energy into mechanical energy. In a D.C. motor, a coil (armature) is placed between the poles of a magnet. When current flows through the coil, the magnetic field exerts a torque on it and it rotates. The commutator reverses the current every half turn, so the torque always acts in the same direction and the coil keeps rotating.

15.7. Describe a simple experiment to demonstrate that a changing magnetic field can induce e.m.f. in a circuit.

Connect a coil to a sensitive galvanometer and push a bar magnet into the coil. The galvanometer deflects while the magnet moves, showing an induced e.m.f., and returns to zero when the magnet stops. Pulling the magnet out deflects the needle the other way, and moving it faster gives a larger deflection. This proves that a changing magnetic field induces e.m.f.

15.8. What are the factors which affect the magnitude of the e.m.f. induced in a circuit by a changing magnetic field?

The induced e.m.f. depends on the rate of change of magnetic flux: faster movement of the magnet or coil gives larger e.m.f. It also depends

on the strength of the magnet, the number of turns in the coil, and the area of the coil. Increasing any of these increases the induced e.m.f., as stated by Faraday's law.

15.9. Describe the direction of an induced e.m.f. in a circuit. How does this phenomenon relate to conservation of energy?

Lenz's law states that the direction of an induced current is always such that it opposes the change that produced it. For example, pushing a magnet into a coil induces a current whose magnetic field repels the magnet, so work must be done against this repulsion. That work becomes the electrical energy of the induced current, so energy is conserved and cannot be created from nothing.

15.10. Draw a labelled diagram to illustrate the structure and working of A.C generator.

Draw a rectangular coil between the poles of a magnet, with its ends connected to two slip rings and brushes leading to an external circuit. As the coil rotates, the magnetic flux through it changes, inducing an alternating e.m.f. that reverses direction every half turn. Label the coil, magnet poles, slip rings, brushes and load.

15.11. What do you understand by the term mutual induction?

Mutual induction is the production of an induced e.m.f. in one coil by a changing current in a neighbouring coil. The changing current produces a changing magnetic field, and this changing flux links with the second coil, inducing an e.m.f. in it. It is the principle on which transformers work.

15.12. What is a transformer? Explain the working of a transformer in connection with mutual induction.

A transformer is a device that changes alternating voltage from one value to another without changing frequency. It has a primary coil and a secondary coil wound on a soft iron core. Alternating current in the primary produces a changing magnetic flux in the core, which links with the secondary coil and induces an alternating voltage there by mutual induction. The voltage ratio equals the turns ratio, Vs/Vp = Ns/Np.

15.13. The voltage chosen for the transmission of electrical power over large distances is many times greater than the voltage of the domestic supply. State two reasons why electrical power is transmitted at high voltage.

Power is transmitted at high voltage for two reasons. First, for a given power, higher voltage means lower current (P = VI), and lower current means much less heat loss in the wires (loss = I^2 R). Second, lower current allows thinner, cheaper wires to be used and reduces the voltage drop along the line.

15.14. Why is the voltage used for the domestic supply much lower than the voltage at which the power is transmitted?

Domestic supply is kept low, at 220 V, because high voltage is extremely dangerous in homes and would damage ordinary appliances, which are designed for low voltage. High transmission voltage is therefore stepped down by transformers at substations to a safe value before it enters houses.

Conceptual Questions

15.1. Suppose someone handed you three similar iron bars and told you one was not a magnet, but the other two were. How would you find the iron bar that was not a magnet?

Use the repulsion test: only magnets repel each other. Take any two bars and bring their ends together in all combinations. The bar that is never repelled by either of the other two, but only attracted, is the unmagnetized iron bar, because attraction can occur between a magnet and iron, while repulsion occurs only between two magnets.

15.2. Suppose you have a coil of wire and a bar magnet. Describe how you could use them to generate an electric current.

Connect the coil to a galvanometer and move the bar magnet in and out of the coil. The changing magnetic flux induces an e.m.f. in the coil and a current flows, shown by the galvanometer deflection. Moving the magnet faster, or using a coil with more turns, gives a larger current.

15.3. Which device is used for converting electrical energy into mechanical energy?

An electric motor converts electrical energy into mechanical energy. Examples are the motors in fans, washing machines, electric trains and water pumps, where current in a coil in a magnetic field produces rotation.

15.4. Suppose we hang a loop of wire so that it can swing easily. If we now put a magnet into the coil, the coil will start swinging. Which way will it swing relative to the magnet, and why?

The coil swings away from the approaching magnet. By Lenz's law, the current induced in the coil produces a magnetic field that opposes the motion of the magnet, so the coil is repelled. The coil therefore moves away from the magnet that is being pushed towards it.

15.5. A conductor wire generates a voltage while moving through a magnetic field. In what direction should the wire be moved, relative to the field to generate the maximum voltage?

The wire should be moved at right angles to the magnetic field, cutting the field lines as fast as possible. The induced e.m.f. is greatest when the motion is perpendicular to the field, and zero when the wire moves parallel to the field lines without cutting them.

15.6. What is the difference between a generator and a motor?

A generator converts mechanical energy into electrical energy: rotating a coil in a magnetic field induces an e.m.f. A motor does the reverse: it converts electrical energy into mechanical energy, as current in a coil in a magnetic field produces rotation. Both use the same electromagnetic principles in opposite directions.

15.7. What reverses the direction of electric current in the armature coil of D.C motor?

The commutator (split ring) reverses the current in the armature coil every half turn. As the coil rotates, each half of the split ring changes contact from one brush to the other, so the current through the coil reverses direction just as the coil passes the vertical position, keeping the torque in the same direction.

15.8. A wire lying perpendicular to an external magnetic field carries current in the direction shown in the diagram. In what direction will the wire move due to the resulting magnetic force?

Apply Fleming's left-hand rule: with the field pointing from N to S (left to right in the diagram) and the current flowing as shown, the thumb points in the direction of the force. The wire therefore moves at right angles to both the field and the current, in the direction given by the left-hand rule for the diagram's field and current directions.

15.9. Can a transformer operate on direct current?

No, a transformer cannot work on direct current. It needs a changing magnetic flux to induce a voltage in the secondary coil, and direct current produces a steady, unchanging flux once switched on. Only alternating current, which continually changes, produces the changing flux needed for mutual induction.

Numerical Problems

15.1. A transformer is needed to convert a mains 240 V supply into a 12 V supply. If there are 2000 turns on the primary coil, then find the number of turns on the secondary coil.

Step 1: Vs/Vp = Ns/Np, so Ns = Np x Vs/Vp.

Step 2: Ns = 2000 x 12/240 = 2000 x 0.05 = 100.

Final answer: Secondary turns = 100.

15.2. A step-up transformer has a turn ratio of 1:100. An alternating

supply of 20 V is connected across the primary coil. What is the secondary voltage?

Step 1: Vs/Vp = Ns/Np = 100/1 = 100.

Step 2: Vs = 100 x Vp = 100 x 20 = 2000 V.

Final answer: Secondary voltage = 2000 V.

15.3. A step-down transformer has a turns ratio of 100:1. An a.c. voltage of amplitude 170 V is applied to the primary. If the current in the primary is 1.0 mA, what is the current in the secondary?

Step 1: For an ideal transformer, Vp x Ip = Vs x Is, and Vs = Vp x (1/100) = 1.7 V.

Step 2: Is = (Vp x Ip) / Vs = (170 x 0.001) / 1.7 = 0.17 / 1.7 = 0.1 A.

Final answer: Secondary current = 0.1 A.

15.4. A transformer, designed to convert the voltage from 240 V a.c. mains to 12 V, has 4000 turns on the primary coil. How many turns should be on the secondary coil? If the transformer were 100% efficient, what current would flow through the primary coil when the current in the secondary coil was 0.4 A?

Step 1: Ns = Np x Vs/Vp = 4000 x 12/240 = 200 turns.

Step 2: For 100% efficiency, Vp x Ip = Vs x Is, so Ip = (Vs x Is)/Vp = (12 x 0.4)/240 = 4.8/240 = 0.02 A.

Final answer: Secondary turns = 200; primary current = 0.02 A.

15.5. A power station generates 500 MW of electrical power which is fed

to a transmission line. What current would flow in the transmission line, if the input voltage is 250 kV?

Step 1: P = V x I, so I = P/V.

Step 2: P = 500 MW = 500 x 10^6 W; V = 250 kV = 250 x 10^3 V.

Step 3: I = (500 x 10^6) / (250 x 10^3) = 2000 A = 2 x 10^3 A.

Final answer: Current = 2 x 10^3 A.

Unit 16: Basic Electronics

Multiple Choice Questions

i. The process by which electrons are emitted by a hot metal surface is known as

(a) boiling

(b) evaporation

(c) conduction

✓ (d) thermionic emission

ii. The particles emitted from a hot cathode surface are

(a) positive ions

(b) negative ions

(c) protons

✓ (d) electrons

iii. The logical operation performed by the gate shown in the book (two

inputs A, B and one output X) is

✓ (a) AND

(b) NOR

(c) NAND

(d) OR

The gate in the book's diagram is the AND gate, whose output is 1 only when both inputs are 1.

iv. AND gate can be formed by using two

(a) NOT gates

(b) OR gates

(c) NOR gates

✓ (d) NAND gates

v. The output of a two-input NOR gate is 1 when:

(a) A is 1 and B is 0

(b) A is 0 and B is 1

✓ (c) both A and B are 0

(d) both A and B are 1

vi. If X = A.B, then X is 1 when:

✓ (a) A and B are 1

(b) A or B is 0

(c) A is 0 and B is 1

(d) A is 1 and B is 0

vii. The output of a NAND gate is 0 when

(a) both of its inputs are 0

(b) both of its inputs are 1

(c) any of its inputs is 0

(d) any of its inputs is 1

Review Questions

16.1. Describe, using one simple diagram in each case, what happens when a narrow beam of electrons is passed through (a) a uniform electric field (b) a uniform magnetic field. What do these results indicate about the charge on electron?

In a uniform electric field, the electron beam bends towards the positive plate, showing that electrons carry negative charge. In a uniform magnetic field, the beam bends in a circular path at right angles to both its motion and the field. Both deflections together confirm that cathode rays are streams of negatively charged particles.

16.2. Explain the working of different parts of oscilloscope.

An oscilloscope (CRO) has an electron gun that produces a narrow beam of electrons by thermionic emission, focusing anodes that sharpen the beam, deflecting plates that move the beam vertically and horizontally according to the input voltage, and a fluorescent screen that glows where the beam strikes. The beam traces the waveform of the input signal on the screen.

16.3. Name some uses of oscilloscope.

An oscilloscope is used to display and measure alternating voltages and waveforms, to find the frequency and amplitude of a signal, to study

sound waves and heartbeats, to test radio and television circuits, and to observe signals that change too fast for an ordinary meter to follow.

16.4. Considering an oscilloscope explain: (i) How the filament is heated? (ii) Why the filament is heated? (iii) Why the anode potential is kept positive with respect to the cathode potential? (iv) Why a large potential is applied between anode and cathode? (v) Why the tube is evacuated?

(i) The filament is heated by passing an electric current through it. (ii) Heating causes thermionic emission, releasing electrons from the cathode. (iii) The anode is kept positive so that it attracts the negatively charged electrons towards the screen. (iv) A large potential gives the electrons high speed, making a bright, sharp spot on the screen. (v) The tube is evacuated so the electrons do not collide with air molecules on their way to the screen.

16.5. What is electron gun? Describe the process of thermionic emission.

An electron gun is the part of a cathode ray tube that produces and focuses a beam of electrons. In thermionic emission, a metal cathode is heated until its free electrons gain enough energy to escape from the metal surface. These emitted electrons are then accelerated by a positive anode to form a beam.

16.6. What do you understand by digital and analogue quantities?

An analogue quantity changes continuously over a range, for example temperature, sound loudness or the position of a clock hand. A digital quantity changes in separate steps and takes only definite values, for example the numbers on a digital clock or the on/off states 0 and 1 in a computer.

16.7. Differentiate between analogue electronics and digital electronics. Write down names of five analogue and five digital devices that are commonly used in everyday life.

Analogue electronics deals with signals that vary continuously, while digital electronics deals with signals that have only two states, 0 and 1. Five analogue devices are: a mercury thermometer, a moving-coil voltmeter, a vinyl record player, an analogue wristwatch and a microphone. Five digital devices are: a digital watch, a calculator, a computer, a mobile phone and a digital camera.

16.8. State and explain for each case whether the information given by the following devices is in analogue or a digital form. a. a moving-coil voltmeter measuring the e.m.f of a cell. b. a microphone generating an electric current. c. a central heating thermostat controlling the water pump. d. automatic traffic lights controlling the flow of traffic.

(a) Analogue: the needle moves continuously with the voltage. (b) Analogue: the current varies continuously with the sound pressure. (c) Digital: the thermostat switches the pump either on or off at a set temperature. (d) Digital: the lights are in one of a few fixed states (red, yellow, green) changed in steps.

16.9. Write down some benefits of using digital electronics over analogue electronics.

Digital electronics is more accurate because values are stored as exact numbers, less affected by noise because only two states must be distinguished, easier to store and copy without loss of quality, cheaper to build with integrated circuits, and able to process, transmit and display information very quickly.

16.10. What are the three universal Logic Gates? Give their symbols and truth tables.

The three universal logic gates are NAND, NOR and NOT-combined gates: NAND and NOR are called universal because any other logic function can be built from NAND gates alone or from NOR gates alone. NAND: output is 0 only when both inputs are 1, otherwise 1. NOR: output is 1 only when both inputs are 0, otherwise 0. NOT: output is the opposite of the input, 1 becomes 0 and 0 becomes 1.

Conceptual Questions

16.1. Name two factors which can enhance thermionic emission.

Thermionic emission is enhanced by increasing the temperature of the cathode, which gives more electrons enough energy to escape, and by using a cathode material with a low work function, such as coated metals, from which electrons escape more easily.

16.2. Give three reasons to support the evidence that cathode rays are negatively charged electrons.

Cathode rays bend towards a positively charged plate in an electric field, showing they are negative. They bend in a magnetic field in the direction expected for negative charges. They also carry the same charge-to-mass ratio wherever they are produced, showing they are identical particles, the electrons.

16.3. When electrons pass through two parallel plates having opposite charges, they are deflected towards the positively charged plate. What

important characteristic of the electron can be inferred from this?

The deflection towards the positive plate shows that the electron carries a negative electric charge, because a negative charge is attracted to a positive plate and repelled by a negative plate. This is direct evidence of the negative charge of the electron.

16.4. When a moving electron enters the magnetic field, it is deflected from its straight path. Name two factors which can enhance electron deflection.

Electron deflection in a magnetic field is increased by a stronger magnetic field and by a higher speed of the electron (up to the point where it leaves the field). A longer region of field also gives the electron more time to bend, increasing the total deflection.

16.5. How can you compare the logic operation X = A.B with usual operation of multiplication?

The logic operation X = A.B behaves exactly like ordinary multiplication of 0 and 1: 0 x 0 = 0, 0 x 1 = 0, 1 x 0 = 0, and 1 x 1 = 1. The output is 1 only when both inputs are 1, just as a product is non-zero only when both factors are non-zero.

16.6. NAND gate is the reciprocal of AND gate. Discuss.

A NAND gate is an AND gate followed by a NOT gate, so its output is the opposite (complement) of the AND output. Where AND gives 1 only when both inputs are 1, NAND gives 0 in that case and 1 in all other cases. NAND is therefore the inverted, or reciprocal, form of AND.

16.7. Show that the circuit given below acts as OR gate.

The circuit in the book combines gates so that its output is 1 whenever either input A or input B is 1, and 0 only when both are 0. Writing the truth table for the combination gives: 0,0 gives 0; 0,1 gives 1; 1,0 gives 1; 1,1 gives 1. This is exactly the truth table of an OR gate, so the circuit acts as an OR gate.

16.8. Show that the circuit given below acts as AND gate.

The circuit in the book combines gates so that its output is 1 only when both inputs A and B are 1, and 0 in all other cases. Its truth table is: 0,0 gives 0; 0,1 gives 0; 1,0 gives 0; 1,1 gives 1. This is exactly the truth table of an AND gate, so the circuit acts as an AND gate.

Unit 17: Information and Communication Technology

Multiple Choice Questions

i. In computer terminology information means

(a) any data

(b) raw data

✓ (c) processed data

(d) large data

ii. Which is the most suitable means of reliable continuous communication between an orbiting satellite and Earth?

✓ (a) microwaves

(b) radiowaves

(c) sound waves

(d) any light wave

iii. The basic operations performed by a computer are

(a) arithmetic operations

(b) non-arithmetic operations

(c) logical operations

✓ (d) both (a) and (c)

iv. The brain of any computer system is

(a) monitor

(b) memory

✓ (c) CPU

(d) control unit

v. Which of the following is not processing?

(a) arranging

(b) manipulating

(c) calculating

✓ (d) gathering

Gathering is collecting data, which happens before processing. Arranging, manipulating and calculating are all forms of processing data into information.

vi. From which of the following we can get information almost about everything?

(a) book

(b) teacher

(c) computer

✓ (d) internet

vii. What does the term e-mail stand for?

(a) emergency mail

✓ (b) electronic mail

(c) extra mail

(d) external mail

Review Questions

17.1. What is difference between data and information?

Data are raw, unorganized facts and figures, such as a list of marks, which by themselves have no clear meaning. Information is data that has been processed, organized and presented in a useful form, such as a result card showing each student's total and position, from which decisions can be made.

17.2. What do you understand by Information and Communication Technology (ICT)?

Information and Communication Technology is the use of computers, telecommunication and other electronic systems to store, process, retrieve and communicate information. It includes hardware, software, networks, the internet, mobile phones and broadcasting, all used to handle information quickly and over long distances.

17.3. What are the components of information technology? Clearly

indicate the function of each component.

The main components are: input devices (keyboard, mouse, scanner) that feed data into the computer; the CPU that processes data; memory that stores data and programs; output devices (monitor, printer) that present information; storage devices (hard disk, flash drive) that keep data for later use; and communication devices (modem, network card) that send and receive data.

17.4. Differentiate between the primary memory and the secondary memory.

Primary memory (RAM and ROM) works directly with the CPU, is fast, and is limited in size; RAM is temporary and loses its contents when power is off, while ROM keeps its contents. Secondary memory (hard disk, flash drive, DVD) is slower but much larger and permanent, keeping data even when the power is off.

17.5. Name different information storage devices and describe their uses.

Storage devices include the hard disk, which stores large amounts of data inside a computer; the flash drive (USB), which carries data portably between computers; the compact disc and DVD, which store music, video and software; the memory card, used in cameras and phones; and magnetic tape, used for backing up large amounts of data.

17.6. Explain briefly the transmission of radiowaves through space.

Radiowaves are electromagnetic waves produced by alternating currents in an antenna. They travel through space at the speed of light without needing a medium, are reflected by the ionosphere for long-distance short-wave broadcasts, or travel in straight lines for television and

satellite links. A receiving antenna converts the waves back into electrical signals.

17.7. How are light signals sent through optical fibre?

Light signals are sent through an optical fibre by total internal reflection. A laser or LED flashes coded pulses of light into one end of the thin glass fibre, and the light reflects repeatedly from the inner walls without escaping, following the fibre's path until it reaches a detector at the far end, which converts it back into an electrical signal.

17.8. What is computer? What is the role of computer in everyday life?

A computer is an electronic machine that accepts data, processes it according to instructions, and gives out information. In everyday life it is used for education, business, banking, communication, entertainment, medicine, transport and government, making work faster, more accurate and easier to store and share.

17.9. What is the difference between hardware and software? Name different softwares.

Hardware is the physical part of a computer that can be touched, such as the keyboard, monitor, CPU and printer. Software is the set of programs and instructions that tells the hardware what to do and cannot be touched. Different softwares are: operating systems (Windows, Android), word processors, spreadsheets, web browsers, and media players.

17.10. What do you understand by the term word processing and data managing?

Word processing is the use of a computer program to create, edit, format,

store and print documents, such as letters and reports. Data managing is the organizing, storing, updating and retrieving of large collections of data, usually in a database, so that information can be found, sorted and used quickly and accurately.

17.11. What is Internet? Internet is a useful source of knowledge and information. Discuss.

The internet is a worldwide network of millions of interconnected computers that share information using common rules. It is a vast source of knowledge because it gives instant access to books, research, news, courses, maps and expert advice from around the world, and allows people to learn, communicate and work together across distances.

17.12. Discuss the role of information technology in school education.

Information technology in schools provides digital textbooks, videos and simulations that make learning clearer, allows students to research topics online, helps teachers prepare lessons and keep records, enables online classes and tests, and develops computer skills that students need for further study and work.

Conceptual Questions

17.1. Why is optical fibre a more useful tool for the communication process?

Optical fibre is more useful because light signals in a fibre suffer very little loss and interference, can carry far more information at higher speed than copper wires, are light and thin, and are secure because the

signal is hard to tap. One fibre can therefore carry thousands of telephone calls or television channels at once.

17.2. Which is more reliable, floppy disk or a hard disk?

A hard disk is more reliable than a floppy disk. It is sealed inside a protective case, stores much more data, reads and writes faster, and is less easily damaged by dust, bending or magnetic fields. Floppy disks are fragile, hold very little data, and lose data easily.

17.3. What is the difference between RAM and ROM memories?

RAM (Random Access Memory) is temporary, fast memory that the CPU uses while working; its contents are lost when the computer is switched off, and data can be both read and written. ROM (Read Only Memory) is permanent memory that keeps its contents without power; it holds the start-up instructions of the computer and normally cannot be changed by the user.

Unit 18: Atomic and Nuclear Physics

Multiple Choice Questions

i. Isotopes are atoms of same element with different

✓ (a) atomic mass

(b) atomic number

(c) number of protons

(d) number of electrons

ii. One of the isotopes of uranium is uranium-238 with atomic number 92. The number of neutrons in this isotope is

(a) 92

✓ (b) 146

(c) 238

(d) 330

Neutrons = mass number - atomic number = 238 - 92 = 146.

iii. Which among the following radiations has more penetrating power?

(a) a beta particle

✓ (b) a gamma ray

(c) an alpha particle

(d) all have the same penetrating ability

iv. What happens to the atomic number of an element which emits one alpha particle?

(a) increases by 1

(b) stays the same

✓ (c) decreases by 2

(d) decreases by 1

v. The half-life of a certain isotope is 1 day. What is the quantity of the isotope after 2 days?

(a) one-half

✓ (b) one-quarter

(c) one-eighth

(d) none of these

After 1 day (one half-life) half remains; after 2 days (two half-lives) half of

a half, or one-quarter, remains.

vi. When Uranium (92 protons) ejects a beta particle, how many protons will be in the remaining nucleus?

(a) 89 protons

(b) 90 protons

(c) 91 protons

✓ (d) 93 protons

In beta decay a neutron changes into a proton, so the atomic number increases by 1: 92 + 1 = 93 protons.

vii. Release of energy by the Sun is due to

(a) nuclear fission

✓ (b) nuclear fusion

(c) burning of gases

(d) chemical reaction

viii. When a heavy nucleus splits into two lighter nuclei, the process would

✓ (a) release nuclear energy

(b) absorb nuclear energy

(c) release chemical energy

(d) absorb chemical energy

ix. The reason carbon-dating works is that

(a) plants and animals are such strong emitters of carbon-14

✓ (b) after a plant or animal dies, it stops taking in fresh carbon-14

(c) there is so much non-radioactive carbon dioxide in the air

(d) when plants or animals die, they absorb fresh carbon-14

Review Questions

18.1. What is difference between atomic number and atomic mass number? Give a symbolical representation of a nuclide.

The atomic number (Z) is the number of protons in the nucleus, which decides which element it is. The mass number (A) is the total number of protons and neutrons in the nucleus. A nuclide is written as A over Z before the element symbol, for example carbon-14 is written with mass number 14 and atomic number 6.

18.2. What do you mean by the term radioactivity? Why are some elements radioactive but some are not?

Radioactivity is the spontaneous emission of radiations (alpha, beta or gamma) by unstable nuclei as they change into more stable nuclei. Some elements are radioactive because their nuclei have an unstable balance of protons and neutrons, usually too many neutrons, while stable elements have a balanced nucleus that does not need to decay.

18.3. How can we make radioactive elements artificially? Describe with a suitable example.

Radioactive elements can be made artificially by bombarding stable nuclei with neutrons or other particles in a nuclear reactor or accelerator. For example, bombarding stable cobalt-59 with neutrons gives cobalt-60, which is radioactive and is used in medicine and industry.

18.4. What are the three basic radioactive decay processes and how do they differ from each other?

The three processes are alpha decay, beta decay and gamma decay. In alpha decay the nucleus emits a helium nucleus (2 protons, 2 neutrons), so mass number falls by 4 and atomic number by 2. In beta decay a neutron becomes a proton and an electron is emitted, so atomic number rises by 1 with no change in mass number. In gamma decay the nucleus releases energy as a gamma photon, with no change in protons or neutrons.

18.5. Write the alpha decay process for protactinium-234 (atomic number 91). Identify the parent and daughter nuclei in this decay.

Protactinium-234 decays by emitting an alpha particle (mass 4, atomic number 2). The daughter nucleus has mass number 234 - 4 = 230 and atomic number 91 - 2 = 89, which is actinium-230. The parent nucleus is protactinium-234 and the daughter nucleus is actinium-230.

18.6. Explain whether the atomic number can increase during nuclear decay. Support your answer with an example.

Yes, the atomic number can increase during beta decay. In beta decay a neutron in the nucleus changes into a proton, so the number of protons rises by one. For example, carbon-14 (6 protons) decays by beta emission into nitrogen-14 (7 protons), so the atomic number increases from 6 to 7.

18.7. What do you understand by half-life of a radioactive element?

The half-life of a radioactive element is the time taken for half of the atoms in a sample to decay. After one half-life half the original amount

remains, after two half-lives one-quarter remains, and so on. Half-life is a fixed property of each isotope and is not affected by temperature, pressure or chemical state.

18.8. Is radioactivity a spontaneous process? Elaborate your answer with a simple experiment.

Yes, radioactivity is a spontaneous process: it happens by itself without any outside cause and cannot be started, stopped or speeded up by ordinary means. In an experiment, a radioactive source placed near a Geiger counter gives counts even in a dark, sealed box with no heating or pressure applied, showing that decay goes on by itself at its own rate.

18.9. What is meant by background radiations? Enlist some sources of background radiations.

Background radiations are the low level of nuclear radiations present everywhere all the time. Their sources are cosmic rays from space, radioactive rocks and soil (such as uranium and radon gas), building materials, food and water containing small amounts of radioactive potassium and carbon, and medical X-rays.

18.10. Describe two uses of radioisotopes in medicine, industry or research.

In medicine, iodine-131 is used to treat thyroid disease and technetium-99m is used for scans of the brain, thyroid, liver and kidneys. In industry, cobalt-60 gamma rays are used to sterilize medical equipment and food, and to check welds in pipes by radiography.

18.11. What are two common radiation hazards? Briefly describe the

precautions that are taken against them.

Two hazards are damage to living cells, which can cause burns, sickness and cancer, and contamination of food, water and surfaces by radioactive material. Precautions are: keep the exposure time short, stay as far as possible from the source, use shielding such as lead screens, wear protective clothing and gloves, and never eat or drink near radioactive sources.

18.12. Complete this nuclear reaction: uranium-235 splits into xenon-140 (atomic number 54) + ? + 2 neutrons. Does this reaction involve fission or fusion? Justify your answer.

Balancing mass numbers: 235 = 140 + mass of unknown + 2, so the unknown mass number is 93. Balancing atomic numbers: 92 = 54 + atomic number of unknown, so it is 38, which is strontium-93. This is fission, because one heavy nucleus (uranium-235) splits into two lighter nuclei with the release of neutrons and energy.

18.13. Nuclear fusion reaction is more reliable and sustainable source of energy than nuclear fission chain reaction. Justify this statement with plausible arguments.

Fusion uses light nuclei such as hydrogen, which are abundant in seawater, so fuel is almost unlimited, while fission fuel (uranium) is limited. Fusion produces mainly harmless helium and very little long-lived radioactive waste, whereas fission produces dangerous waste that lasts thousands of years. Fusion is also safer because it stops by itself if conditions fail, while a fission chain reaction must be controlled to avoid accidents.

18.14. A nitrogen nuclide (nitrogen-14, atomic number 7) decays to become an oxygen nuclide by emitting an electron. Show this process with an equation.

Emitting an electron is beta decay, in which a neutron becomes a proton: mass number stays 14 while atomic number rises from 7 to 8. The equation is: nitrogen-14 changes into oxygen-14 plus an electron (beta particle). The parent is nitrogen-14 and the daughter is oxygen-14.

18.15. Determine which of these radioactive decay processes are possible: (a) polonium-214 changes into polonium-214 + alpha particle (b) thorium-230 changes into radium-226 + alpha particle (c) protactinium-233 changes into uranium-233 + beta particle (d) carbon-14 changes into nitrogen-14 + beta particle

(a) Not possible: mass and charge are not balanced, because an alpha particle is emitted without any change in the nucleus. (b) Possible: mass 230 = 226 + 4 and atomic number 90 = 88 + 2, both balanced. (c) Possible: beta decay keeps mass at 233 and raises atomic number from 91 to 92, which is exactly protactinium-233 changing into uranium-233. (d) Possible: carbon-14 (atomic number 6) decays to nitrogen-14 (atomic number 7) by beta emission, mass unchanged and atomic number increased by 1.

Conceptual Questions

18.1. Is it possible for an element to have different types of atoms? Explain.

Yes, an element can have different types of atoms called isotopes.

Isotopes have the same number of protons (same atomic number) but different numbers of neutrons, so they have different mass numbers. For example, carbon has carbon-12 and carbon-14, both with 6 protons but with 6 and 8 neutrons respectively.

18.2. What nuclear reaction would release more energy, the fission reaction or the fusion reaction? Explain.

Per unit mass of fuel, fusion releases more energy than fission. Fusing light nuclei into helium converts a larger fraction of mass into energy than splitting heavy nuclei, because the binding energy per nucleon is greatest near iron and the gain from light nuclei is larger. That is why the Sun's fusion gives out enormous energy from a small mass of hydrogen.

18.3. Which has more penetrating power, an alpha particle or a gamma ray photon?

A gamma ray photon has far more penetrating power than an alpha particle. Alpha particles are heavy and charged, so they lose energy quickly and are stopped by paper or a few centimetres of air. Gamma rays have no charge or mass, so they pass through many centimetres of lead and can only be weakened, not fully stopped, by thick shielding.

18.4. What is the difference between natural and artificial radioactivity?

Natural radioactivity is the decay of unstable nuclei that occur in nature, such as uranium, radium and carbon-14, without human action. Artificial radioactivity is produced by humans by bombarding stable nuclei with particles in reactors or accelerators, creating new radioactive isotopes such as cobalt-60 and iodine-131 that do not occur naturally in large amounts.

18.5. How long would you likely have to wait to watch any sample of radioactive atoms completely decay?

In theory, you would have to wait an infinite time, because after each half-life half of what remains still survives. In practice, after about ten half-lives less than one-thousandth remains, which is usually too small to detect, so the sample is treated as fully decayed even though a few atoms may still remain.

18.6. Which type of natural radioactivity leaves the number of protons and the number of neutrons in the nucleus unchanged?

Gamma decay leaves both the number of protons and neutrons unchanged. The nucleus only loses energy by emitting a gamma photon and drops from an excited state to a lower energy state, with no change in mass number or atomic number.

18.7. How much of a 1 g sample of pure radioactive substance would be left undecayed after four half-lives?

After one half-life 1/2 g remains, after two 1/4 g, after three 1/8 g, and after four half-lives 1/16 g remains. So 1/16 of 1 g = 0.0625 g would be left undecayed after four half-lives.

18.8. Tritium (hydrogen-3) is a radioactive isotope of hydrogen. It decays by emitting an electron. What is the daughter nucleus?

Tritium decays by beta emission: a neutron becomes a proton, so atomic number rises from 1 to 2 while mass number stays 3. The daughter nucleus is helium-3, with 2 protons and 1 neutron.

18.9. What information about the structure of the nitrogen atom can be

obtained from its nuclide nitrogen-14 (atomic number 7)? In what way is the atom in nitrogen-14 different from the atom in nitrogen-16?

Nitrogen-14 has 7 protons and 14 - 7 = 7 neutrons, with 7 electrons in a neutral atom. Nitrogen-16 has the same 7 protons and 7 electrons but 16 - 7 = 9 neutrons, so it is a different isotope of nitrogen with two extra neutrons and a larger mass.

Numerical Problems

18.1. The half-life of nitrogen-16 is 7.3 s. A sample of this nuclide of nitrogen is observed for 29.2 s. Calculate the fraction of the original radioactive isotope remaining after this time.

Step 1: Number of half-lives = time / half-life = 29.2 / 7.3 = 4.

Step 2: Fraction remaining = (1/2)^4 = 1/16.

Final answer: Fraction remaining = 1/16.

18.2. Cobalt-60 is a radioactive element with half-life of 5.25 years. What fraction of the original sample will be left after 26 years?

Step 1: Number of half-lives = 26 / 5.25 = 4.95, about 5.

Step 2: Fraction remaining = (1/2)^5 = 1/32.

Final answer: Fraction remaining = 1/32 (book answer).

18.3. Carbon-14 has a half-life of 5730 years. How long will it take for the quantity of carbon-14 in a sample to drop to one-eighth of the initial quantity?

Step 1: One-eighth = (1/2)^3, so 3 half-lives are needed.

Step 2: Time = 3 x 5730 = 17,190 years = 1.72 x 10^4 years.

Final answer: Time = 1.72 x 10^4 years.

18.4. Technetium-99m is a radioactive element used to diagnose brain, thyroid, liver and kidney diseases. This element has half-life of 6 hours. If there is 200 mg of this technetium present, how much will be left in 36 hours?

Step 1: Number of half-lives = 36 / 6 = 6.

Step 2: Amount remaining = 200 x (1/2)^6 = 200 / 64 = 3.125 mg.

Final answer: Amount left = 3.12 mg.

18.5. Half-life of a radioactive element is 10 minutes. If the initial count rate is 368 counts per minute, find the time for which count rate reaches 23 counts per minute.

Step 1: 368 to 23 is a factor of 368/23 = 16 = 2^4, so 4 half-lives pass.

Step 2: Time = 4 x 10 = 40 minutes.

Final answer: Time = 40 minutes.

18.6. In an experiment to measure the half-life of a radioactive element, the following results were obtained: count rate per minute 400, 200, 100, 50, 25 at times 0, 2, 4, 6, 8 minutes. Plot a graph between the count rate and time in minutes. Measure the value for the half-life of the element from the graph.

Step 1: Plot count rate on the vertical axis against time on the horizontal

axis; the points form a falling curve.

Step 2: The count rate halves from 400 to 200 in 2 minutes, from 200 to 100 in the next 2 minutes, and so on.

Step 3: The time for the count rate to fall to half its value is therefore 2 minutes at every step.

Final answer: Half-life = 2 minutes.

18.7. A sample of certain radioactive element has a half-life of 1500 years. If it has an activity of 32000 counts per hour at the present time, plot a graph of the activity of this sample over the period in which it will reduce to 1/16 of its present value.

Step 1: 1/16 = (1/2)^4, so 4 half-lives are needed: 4 x 1500 = 6000 years.

Step 2: Activities: at 0 years 32000; at 1500 years 16000; at 3000 years 8000; at 4500 years 4000; at 6000 years 2000 counts per hour.

Step 3: Plot these points with time on the horizontal axis and activity on the vertical axis, joining them with a smooth falling curve.

Final answer: Activity falls to 2000 counts per hour (1/16 of present) after 6000 years.

18.8. Half-life of a radioactive element was found to be 4000 years. The count rates per minute for 8 successive hours were found to be 270, 280, 300, 310, 285, 290, 305, 312. What does the variation in count rates show? Plot a graph between the count rates and time in hours. Why the graph is a straight line rather than an exponential?

Step 1: The small up-and-down variation (270 to 312) around a mean of

about 295 shows the random nature of radioactive decay.

Step 2: Plotted against time, the points form an almost horizontal straight line, not a falling curve.

Step 3: This is because the half-life (4000 years) is enormously longer than the 8-hour observation period, so the activity hardly falls at all in 8 hours.

Final answer: Variation shows random decay; graph is almost horizontal because the half-life is very long compared with 8 hours.

18.9. Ashes from a campfire deep in a cave show carbon-14 activity of only one-eighth the activity of fresh wood. How long ago was that campfire made?

Step 1: One-eighth = (1/2)^3, so 3 half-lives have passed.

Step 2: Half-life of carbon-14 = 5730 years. Time = 3 x 5730 = 17,190 years.

Final answer: The campfire was made 17,190 years ago.